
Typical: Typed Decisions Read Directly from Pretrained Language Models

Guy Nachshon
OzLabs
Guy@ozlabs.ai

Abstract

Software that asks a language model for a decision – route this ticket, approve this refund, rate this alert – needs a probability distribution over options it names at request time, plus a usable way to say that none of them fit. **Typical** reads that distribution directly out of a pretrained causal language model, without generating a token. The state is encoded once into a key-value cache; each question and its runtime candidates form a short causal suffix; a head at 71% of backbone depth scores the terminal decision state against each candidate’s own contextual hidden state. Output types are contracts on the distribution: categorical Choice, ordinal Score, Bernoulli Noul, and an abstention gate outside the candidate softmax. The contextual readout keeps the question-conditioned signal of the standard letter-logit interface (Δ_q^{sh} .106 vs. .109) with a quarter of its log-odds shift under candidate-set edits. We measure that signal with Δ_q , a shuffled-question control that separates question-conditioned capability from candidate-set priors, and use it to show that the capability lives in computation over the candidates: in all six configurations we tested, a decision state computed before the candidates are seen stays within .03 of zero, against .09–.12 for readouts over them. A training recipe built on counterfactual rubric groups, decision-type balancing, null augmentation and calibration-based checkpoint selection lifts held-out rule families by 35–40 points and cuts held-out ordinal NLL threefold at 14B. A decision costs 45 ms at 1.7B and 59 ms at 14B in a research harness, and 15.5–17 ms on the optimized 1.7B serving path; the 14B model reaches .931 on the JevBench standard tier. A train-render $\times$ eval-render design on 605 held-out long documents shows that evidence-first training adds 11.2 points ($p = 5 \times 10^{-10}$), reproduced at a second seed. We release 1.7B and 4B models and an inference package.

1 Introduction

A growing share of language-model calls in production software ask for a decision rather than a paragraph: which of forty queues a ticket belongs in, whether a transaction is fraud, how severe an alert is, which tool an agent should call next. The caller needs a probability distribution over a label set it defines at request time, and it needs to know when none of the labels fit. The usual route to that distribution goes through text. The model generates an answer symbol or a JSON object, or the caller constrains decoding, or it compares candidate log-probabilities. Each of these inherits the costs of the generative interface: decoding latency, parsing, and a well-documented sensitivity to how the answer is written, including option-order and option-symbol biases [Zheng et al., 2023, Pezeshkpour and Hruschka, 2024] and first-token probabilities that disagree with the text the model goes on to produce [Wang et al., 2024a].

A pretrained model does not need to write its answer down for the decision to exist. When a causal language model reads a question followed by a list of options, the winning option is already decod-

able from its hidden states before the answer symbol is bound [Wong et al., 2026]. **Typical** makes that internal decision the deployed interface. A call has the signature

$$(\text{state}, \text{typed question}, \text{runtime candidate set}) \mapsto (P(a_1), \dots, P(a_K), P(\emptyset)),$$

and nothing is generated or parsed. The state (a ticket, a policy document, an agent trace) is encoded once into a key-value cache. Each question against it is a short causal suffix containing the question and its candidates, and a head at 71% of the backbone’s depth reads the terminal decision state against each candidate’s own contextual hidden state. The output type is a contract on the returned distribution: categorical *Choice*, ordinal *Score*, Bernoulli *Noul*, and an abstention event $\emptyset$ computed by a gate outside the candidate softmax (Figure 1).

Building this interface raised three design questions, and each has a measured answer.

What should the head read? Each candidate’s contextual state. Scoring the decision state against the candidates’ contextual hidden states keeps the question-conditioned signal of the standard letter-logit interface (Δ_q^{sh} .106 vs. .109, within seed noise across two seeds), reduces the shift in pairwise log-odds under candidate-set edits to a quarter, and adds 2–7 points on natural language inference. Scoring the same decision state against an independent embedding of each option’s text loses 13 points of knowledge accuracy (Section 6.1).

Where does question-conditioned capability live? In computation over the candidates. We measure it with Δ_q , the accuracy lost when the real question is replaced by a shuffled one while the state and candidate set stay fixed, which extends the choices-only control of Balepur et al. [2024]. It separates question-conditioned capability from candidate-set priors, a distinction that accuracy and calibration both miss. In every configuration we tested (six factorizations and objectives, two backbone depths), a decision state computed before the candidates are seen stays within .03 of zero, while readouts that process the candidates in context retain .09–.12, matching their listwise teacher (Section 6.2). Reading at 71% of depth yields one model that is strong on both evidence and knowledge; reading the final layer produces a model at the SNLI hypothesis-only baseline (Section 6.3).

How should the output be typed? As contracts that the head and target enforce. Keeping abstention as a gate, and never as a rendered “none of the above” candidate, gives an abstention score that behaves consistently from 5 to 150 candidates and lowers MMLU-Pro false abstention from .692 to .003 (Section 6.4). An ordinal target cuts Score’s held-out NLL from 2.07 to 1.23 at unchanged accuracy, and a Bernoulli head makes yes/no decisions exactly invariant to label order (Section 7.1).

The second half of the paper is about training decision models, and here the data distribution is as strong a lever as the architecture. Counterfactual rubric groups, in which the same state and candidates appear under different rubrics with different gold answers, make the rubric the only way to fit the data and lift held-out families, grammars and rubric styles by 35–40 points. Balancing decision types, augmenting workflow rows with null targets, and selecting checkpoints on held-out likelihood each move a measurable axis (Sections 7.2 and 7.3). In a single-stream research harness a decision costs 45, 57 and 59 ms at 1.7B, 4B and 14B, so an $8\times$ larger backbone is $1.3\times$ slower; the optimized serving path decides in 15.5–17 ms warm at 1.7B. The 14B model reaches .931 on the JevBench standard tier (Sections 7.4 and 7.5).

We also treat rendering, the way the state and options are laid out as text, as an experimental variable. A 2×2 design that crosses training render with evaluation render on 605 held-out long documents measures a data effect that a 19-item benchmark cell could not resolve: training with the evidence placed first adds 11.2 points ($p = 5 \times 10^{-10}$), reproduces at a second seed, and is corroborated at 14B (Section 7.6). The corrected checkpoints are the current public releases, at .947 (1.7B) and .950 (4B) on those documents.

Contributions.

1. **A direct decision interface** for pretrained causal language models: a cached-state decision pass with a contextual-candidate readout at mid-depth, typed output contracts (Choice, Score, Noul) and a factored abstention gate with exact structural properties, returning distributions over runtime candidates without generating tokens (Section 3).
2. **An account of where the decision lives**, built on Δ_q : matched comparisons of readouts, decision-state factorizations, depths and abstention designs, which together fix each component of the interface (Section 6).

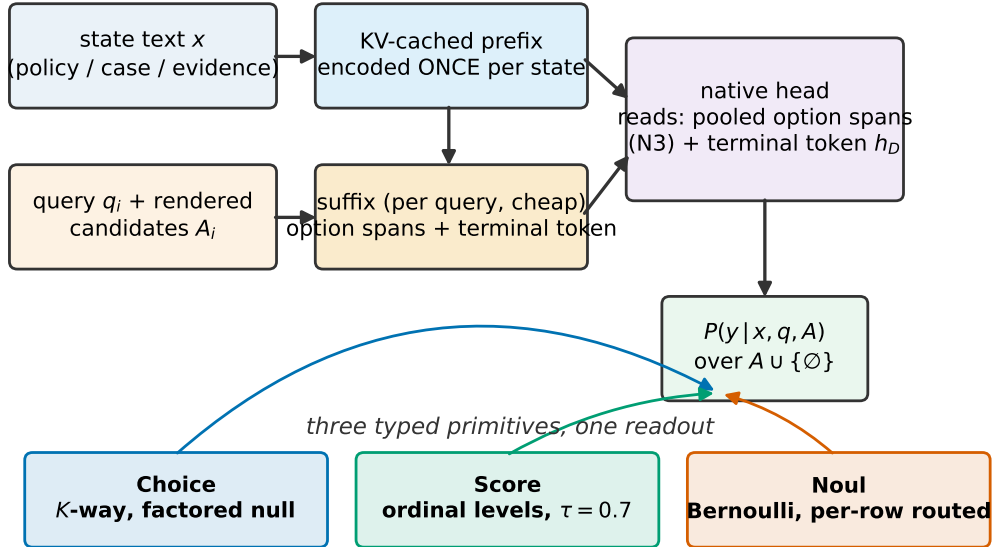


Figure 1: The decision pass. The state is encoded once into a key-value cache. Each question is a short causal suffix holding the question and its runtime candidates; the head reads the terminal decision state h_D against each candidate’s pooled contextual state c_j^{ctx} at 71% of backbone depth. One readout serves three typed contracts (Choice, Score, Noul), and abstention is a separate gate outside the candidate softmax.

3. **A training recipe for decision models:** counterfactual rubric groups, decision-type balancing, a soft-target uncertainty corpus, null augmentation, evidence-first long states and calibration-based selection, each with a matched comparison (Sections 4 and 7).
4. **Render-crossed evaluation**, which separates data effects from interface compatibility and shows how much of a benchmark score rendering alone can move (Sections 7.6 and 7.7).
5. **Open models and code:** typical-small (Qwen3-1.7B) and typical-medium (Qwen3.5-4B), with a self-contained inference package (`pip install typical-ai`) whose outputs match the training-time decider exactly.

2 Related work

Reading decisions from hidden states. Hidden states carry linearly decodable structure [Alain and Bengio, 2016, Hewitt and Liang, 2019, Belinkov, 2022], different depths carry different abstractions [Tenney et al., 2019, Belrose et al., 2023], states anticipate tokens not yet emitted [Pal et al., 2023, Wu et al., 2024], and circuit analysis localizes the step that binds a chosen option to its symbol [Lieberum et al., 2023]. The closest prior work is Wong et al. [2026]: in ordinary multiple-choice prompting the winning option is decodable in content space at the option boundary, before the answer symbol is bound, and Bhatt et al. [2026] trace the same emergence across languages. Typical makes that pre-symbol decision the deployed interface and measures what the interface needs: candidates inside the forward pass (Section 6.2), a readout over each candidate’s contextual state (Section 6.1), and a mid-depth tap, consistent with evidence that upper layers are partly redundant and that early exit is a viable efficiency lever [Gromov et al., 2024, Men et al., 2025, Schwartz et al., 2020].

The answer-symbol interface. Answer symbols bring option-order and symbol-identity biases [Zheng et al., 2023, Pezeshkpour and Hruschka, 2024, Yang et al., 2025b], motivate binding fixes [Xue et al., 2024], and yield first-token probabilities that disagree with the generated text [Wang et al., 2024a]. Constrained decoding produces structured outputs [Geng et al., 2023, Willard and

Louf, 2023] and can cost accuracy [Tam et al., 2024]. The choices-only control of Balepur et al. [2024] measures how much multiple-choice accuracy needs no question. Typical removes the symbol from the readout, and Δ_q (Definition 5.1) extends the choices-only control to a shuffled real question.

Runtime-defined label spaces. Unseen labels have been scored through joint input-label embeddings [Pappas and Henderson, 2019], entailment reformulation [Yin et al., 2019], label-conditioned encoders [Halder et al., 2020], and runtime schemas encoded jointly with the text [Zaratiana et al., 2024, 2025]. Cost follows the cross- versus dual-encoder trade-off: one forward pass per candidate [Nogueira and Cho, 2019], a candidate-independent dot product [Reimers and Gurevych, 2019], or late interaction and decomposition in between [Humeau et al., 2019, Khattab and Zaharia, 2020, Cao et al., 2020a, MacAvaney et al., 2020, de Jong et al., 2023], often with a cross-encoder compiled into a cheaper student [Lu et al., 2022]. Decoder models with a classification head fix the labels as indices [Li et al., 2023]. Typical renders all candidates in one causal suffix over a cached state, so each candidate’s representation carries its slot and its set, and one state prefix serves many decisions. Δ_q measures what candidate-independent factorizations give up (Section 6.2); such a scorer stays useful as a shortlisting front end for large sets (Section 7.5).

Prefix sharing and the decode loop. Prefix sharing across suffixes is established systems work [Juravsky et al., 2024, Zheng et al., 2024, Kwon et al., 2023, Gim et al., 2023, Cheng et al., 2023, Pope et al., 2022, Dao et al., 2022], and speculative decoding and Medusa exist because sequential decode steps are overhead-bound [Leviathan et al., 2022, Chen et al., 2023, Cai et al., 2024]. Typical adds no kernel; it deletes the decode loop and reports the resulting cost from 1.7B to 14B (Section 7.5).

Ordinal targets, abstention and probability quality. Score uses the soft-target member of the ordinal-regression family [McCullagh, 1980, Frank and Hall, 2001, Niu et al., 2016, Cao et al., 2020b, Shi et al., 2023b, Díaz and Marathe, 2019]. The reject option is classical [Chow, 1970, Geifman and El-Yaniv, 2017, 2019], scored by confidence or energy [Hendrycks and Gimpel, 2016, Liu et al., 2020] and benchmarked by out-of-scope intent detection [Larson et al., 2019, Casanueva et al., 2020]; Typical makes abstention a gate outside the candidate softmax with exact algebraic properties (Section 3.6). Temperature scaling repairs much in-distribution miscalibration [Guo et al., 2017], binned ECE is biased [Pakdaman Naeini et al., 2015, Kumar et al., 2019], and proper scoring rules, including the ordinal ranked probability score, are the primary target [Gneiting and Raftery, 2007, BRIER, 1950, Epstein, 1969, MURPHY, 1970]. Label smoothing [Szegedy et al., 2016, Müller et al., 2019] and distillation [Hinton et al., 2015, Kim and Rush, 2016, Yuan et al., 2020, Zhang and Sabuncu, 2020] improve probability quality, context distillation compiles a prompt into weights [Askell et al., 2021, Snell et al., 2022], and fine-tuning can distort calibration [Desai and Durrett, 2020, Wang et al., 2025]. Typical shapes probability quality at training time with soft and ordinal targets, a Brier term and likelihood-based selection (Section 4.6); its distillation teacher is the same backbone under a three-shot rendering.

Fine-tuning and composition. Fine-tuning can distort pretrained features [Kumar et al., 2022], is close to in-context learning under fair comparison [Mosbach et al., 2023], and with LoRA [Hu et al., 2021] learns and forgets less [Biderman et al., 2024] while still forgetting under continual training [Luo et al., 2025]. Compositional benchmarks show where learned grammars stop extending [Lake and Baroni, 2017, Keysers et al., 2019, Kim and Linzen, 2020, Hupkes et al., 2020]. Typical’s counterfactual rubric groups make the rubric the only way to fit the corpus, and its level-7 split is a held-out composition probe (Section 4.2).

Long context and data pipelines. Evidence position, length and distractors change long-context answers [Liu et al., 2024, Levy et al., 2024, Shi et al., 2023a, Hsieh et al., 2024], and pipeline defects propagate silently into models [Sambasivan et al., 2021]. Typical treats evidence placement as a training factor crossed with evaluation render (Sections 4.5 and 7.6).

Single-pass computation. Chain-of-thought adds serial computation that a constant-depth transformer lacks in one pass [Wei et al., 2022, Li et al., 2024, Merrill and Sabharwal, 2023b,a, Feng et al., 2023] and helps mainly on math and symbolic tasks [Sprague et al., 2024]; pause and filler tokens [Goyal et al., 2023, Pfau et al., 2024] and implicit or latent reasoning [Deng et al., 2023, 2024,

Hao et al., 2024] supply that computation without verbalized steps. Typical is a single fixed-depth pass and a natural terminal layer for such methods (Section 8).

Typed decision benchmarks. JevBench targets the interface of closed-weight commercial “System-One” decision models such as TypeSafe’s Jev, documented only by vendor pages and a third-party black-box probe. Its open leaderboard entries publish weights and scores, and only GLiNER2 [Zaratiana et al., 2025] and the ModernBERT backbone of one entry [Warner et al., 2025] have method papers. Typical addresses the same interface with an open architecture, matched ablations and released weights.

3 Typical: the decision pass

Typical turns a pretrained causal language model into a decision function. A call supplies a state, a typed question and a candidate set, and receives a probability for every candidate plus a probability of abstention, read from hidden states in one pass over a short suffix. This section defines the task (Section 3.1), the pass (Section 3.2), the readout (Section 3.3) and the layer it reads (Section 3.4), the typed heads (Section 3.5), the abstention gate (Section 3.6), and the cost model (Section 3.7).

3.1 Typed decisions

Definition 3.1 (Typed decision). A decision is a tuple $D = (x, q, A, t)$: a state x (the document, transcript or trace the decision is about), a question q (the instruction or rubric), a candidate set $A = \{a_1, \dots, a_K\}$ of label strings supplied at request time, and a type $t \in \{\text{CHOICE}, \text{SCORE}, \text{NOUL}\}$. The model returns

$$P(a_j \mid x, q, A), \quad j = 1, \dots, K, \quad P(\emptyset) = P(\emptyset \mid x, q, A), \quad P(\emptyset) + \sum_j P(a_j \mid x, q, A) = 1,$$

and emits no token.

The type is a contract on the returned distribution, and each contract is enforced by the head and the training target. **Choice** is categorical over A . **Score** is categorical over an ordered A (severity levels, priorities), trained so that mass on a level adjacent to the gold level costs less than mass on a distant one. **Noul** is a Bernoulli probability for a single proposition with $A = \{\text{yes}, \text{no}\}$, invariant to the order in which the caller lists the two labels. The abstention event $\emptyset$ is available under every type and is never one of the candidates. Because A is text supplied per call, the model has no output index for any label and must read the candidates to score them.

3.2 Cached state and causal suffix

The state is encoded once as an ordinary causal prefix and its key-value cache is retained. Each question against it is a short causal suffix: the question text, the candidates rendered as lettered lines, an answer cue, and a terminal decision token. The suffix runs through the backbone attending to the cached state, and a batch of suffixes against one state runs in chunks without attending to one another. A decision is therefore exactly a forward pass on $\text{concat}(x, q, A)$ with the state half read from cache. We verify this with an equivalence test on the real model: cached-state decisions match the full-concatenation path to within 10^{-3} across suffix chunk sizes 1, 2, 3, 5 and 64, are consistent under permutation of the queries in a batch, and repeat exactly. No question recomputes the state, and no question sees another.

3.3 The contextual-candidate readout

One forward pass over the suffix yields two things at the tap layer: the terminal decision state h_D , the hidden state at the decision token, and for each candidate a contextual state c_j^{ctx} , the mean of the hidden states over the tokens of candidate j ’s rendered span. Candidate j receives the score

$$u = W_h h_D, \quad v_j = W_c c_j^{\text{ctx}}, \quad s_j = \frac{u^\top v_j}{\sqrt{d_v}} + w^\top (u \odot v_j) + g(u \odot v_j), \quad (1)$$

where W_h and W_c are layer-normalized linear projections to width d_v , w is a vector and g a two-layer MLP. The score is a bilinear form in h_D and c_j^{ctx} with a small learned correction on their elementwise product. The option letters stay in the rendered text and are never read out.

Because the candidates are tokens in the same causal computation as the state and the question, c_j^{ctx} carries the candidate’s meaning together with its slot and the set it appears in, and h_D has attended to all of them. We compare three readouts that differ only in what h_D is scored against (Section 6.1):

- **N1, letter-logit readout.** h_D is scored against the output embedding of each option letter, i.e. the next-token logits of the letters. Nothing is generated. This is the standard multiple-choice interface read at the decision token.
- **N2, candidate-blind semantic readout.** h_D is scored with Equation (1) against an embedding of the candidate text computed independently by a frozen text-embedding model, which carries the label’s meaning and no slot identity.
- **N3, contextual-candidate readout.** h_D is scored with Equation (1) against the span-pooled contextual state c_j^{ctx} . This is the readout Typical deploys.

N3 keeps N1’s question-conditioned signal within seed noise, cuts its pairwise log-odds shift under candidate-set edits to about a quarter, and gains on evidence tasks. N2, with the same scorer, keeps Δ_q^{sh} of .058 against the teacher’s .102 and scores .012 on Banking77 at $K = 77$ (Section 6.1).

3.4 Mid-depth tap

The backbone is truncated at about 71% of its depth: the readout taps layer 20 of 28 at 1.7B, 26 of 36 at 4B and 8B, and 28 of 40 at 14B. The layers above the tap are removed from the model, so the top 8 layers at 1.7B, 10 at 4B and 8B, and 12 at 14B are never computed for any token, state or suffix. LoRA [Hu et al., 2021] at rank 16 adapts the top eight kept layers (layers 13–20 at 1.7B), and everything below is frozen.

The tap is chosen for what the representation holds. Our first full-scale model read the final layer and scored 58.6 on SNLI, the hypothesis-only baseline: it ignored the premise. For the contextual readout, moving the tap from layer 28 to layer 20 with everything else fixed raises SNLI, MNLI, ANLI and BoolQ from .863, .752, .431 and .737 to .906, .865, .519 and .834, and keeps the question-conditioned signal (Δ_q^{sh} .118 against .117), so one model carries both evidence and knowledge (Section 6.3). All released and ladder models use this tap.

3.5 Typed heads

Choice, Score and Noul share the backbone, the tap and the readout of Equation (1). They differ in the training target and, for Noul, the terminal head, and every row is routed by its own type.

Choice is the K -way readout trained with cross-entropy against the gold distribution over $A \cup \{\emptyset\}$, composed through the gate of Section 3.6.

Score uses the Choice readout over the rendered levels with an ordinal-smoothed target. For gold level y ,

$$\tilde{t}_j \propto \exp\left(-\frac{|j - y|}{\tau}\right), \quad \tau = 0.7,$$

normalized over the valid levels; $\tau \rightarrow 0$ recovers the one-hot Choice target. This is a change of target with no new parameters. On held-out urgency it takes NLL from 2.07 to 1.23 at flat accuracy (.505 to .508), and on JevBench’s 12 ordinal items it changes no decision (Section 7.1).

Noul uses a dedicated Bernoulli head on a query-only suffix that renders no candidate text:

$$P(\text{yes}) = \sigma(w_n^\top h_D),$$

with the complement assigned to “no”. Since no candidate enters the computation, reversing the label order cannot change $P(\text{yes})$; a two-way Choice over the same labels moves it by up to .55 under reversal (Section 7.1). The abstention gate applies to Noul unchanged.

Per-row routing. A row takes the Bernoulli path if and only if its candidate set is exactly {yes, no}, and Score rows take the ordinal target; every other row is scored by the Choice path, bit-identical to a model with the typed heads disabled. Routing is a property of the row because a batch mixes types: an early build that applied the Bernoulli head to every row scored Choice and Score rows with no candidate text and fell to chance on them (Appendix A.5).

3.6 Factored abstention gate

Abstention is a separate gate with its own inputs, outside the candidate softmax. Given scores $s_1, \dots, s_K$ from Equation (1), the gate reads a vector z of permutation-invariant statistics over the valid candidates (maximum score, top-2 margin, mean score, log-mean-exp of the scores), the projected decision state, and the mean and variance of the projected candidate states v_j , and computes $r = \sigma(g_\varnothing(z))$. The returned distribution is

$$P(a_j \mid x, q, A) = (1 - r) \frac{\exp(s_j/T)}{\sum_{k=1}^K \exp(s_k/T)}, \quad P(\varnothing) = r, \quad (2)$$

implemented as logits $\ell_j = \log \text{softmax}(s/T)_j + \log(1 - r)$ and $\ell_\varnothing = \log r$, so a standard cross-entropy over $K + 1$ outcomes trains both parts. Two properties follow exactly from the construction.

1. *Pairwise non-null odds are invariant to abstention.* For fixed scores, $P(a_j)/P(a_k) = \exp((s_j - s_k)/T)$, independent of r : the multiplier $(1 - r)$ is common to every non-null candidate, so abstention never reorders or reweights the candidates among themselves. The contextual scores s_j are themselves computed in the context of the whole set and remain set-dependent.
2. *The gate is temperature-invariant.* T enters only the candidate softmax and r is computed from untempered statistics, so recalibrating T sharpens or flattens the candidate distribution without moving $P(\varnothing)$.

Together with the exact order invariance of the Bernoulli head (Section 3.5), these are the structural guarantees of the output layer. The gate is also a symmetric function of the candidate set: every input to z is invariant to permuting the pairs (s_j, v_j) , so it adds no dependence on candidate order beyond what the contextual scores carry. Whether to abstain depends on what was offered, so r is set-dependent by design. The null is never rendered as a candidate line. The matched experiment in Section 6.4 removes the rendered “none of the above” line from a model that already has this gate and shows why: a rendered null competes for softmax mass that dilutes as K grows, and removing it takes MMLU-Pro false abstention from .692 to .003.

3.7 Cost model

M decisions against one state cost one prefix pass and M suffix passes,

$$O(|x|) + M \cdot O(|q| + |A|)$$

token positions, where $|A|$ counts the rendered candidate tokens. Each suffix position attends to the cached state, and every position, in the prefix and in the suffix, runs through only the kept 71% of the layers. The state term is paid once and amortized over every question asked of it; the per-decision term grows with the length of the question and the candidate text, and is independent of the other questions. On one H100 in our single-stream research harness, a decision at $K = 2$ costs 45, 57 and 59 ms at 1.7B, 4B and 14B, so an $8\times$ larger backbone is $1.3\times$ slower. At $K = 256$ the candidate term dominates and grows with the backbone: the marginal per-query cost in a batch of 32 is 28.0 ms at 1.7B and 109.7 ms at 14B, against 2.7 and 3.9 ms at $K = 2$. For large candidate sets a candidate-blind scorer, whose cost does not depend on rendering candidates through the backbone, shortlists to a bounded set for the contextual readout; Section 7.5 gives the full sweep and the crossover.

4 Training decision models

The architecture of Section 3 fixes what a decision model can represent; the training distribution decides what it learns to represent. Six choices define the recipe, and Section 7 measures each. Every row is a typed decision rendered as in Section 3.2, and the loss is cross-entropy against a hard or soft gold distribution over $A \cup \{\varnothing\}$, with a Brier term and, for the 14B model, a distillation term (Section 4.6). Training uses AdamW in bf16 at an effective batch size of 64 for 8,000–12,000 steps (Appendix A.7). Table 1 lists the seven training corpora.

Table 1: Training corpora. “Held out” counts whole families, domains, styles or parameter ranges excluded from training entirely, distinct from a random validation split; Section 5.2 maps these onto generalization axes. No corpus contains any of the 231 public JevBench ids.

Corpus	Family	Train rows	Held out	Role
data_v5	<i>E</i>	633,213	label wordings	Evidence (NLI, BoolQ) and wide-vocabulary classification
data_kb	<i>K</i>	147,447	TruthfulQA-MC1, GPQA	Closed-book multiple-choice knowledge
data_wf	<i>W</i>	72,006	3 of 15 families	Rubric-conditioned workflow (Noul, Choice, Score)
data_wf_hf	<i>W</i>	86,600	3 external test/demo splits	External workflow sets (forms, rules, distillation)
data_wf_long	<i>W</i>	23,318	–	Long-state (multi-paragraph policy) workflow rows
data_wh	<i>W</i>	60,605	3 domains, 2 styles, 1 grammar per level, level 7	DecisionMix v2 counterfactual rubric groups
data_u	<i>U</i>	31,029	4 synthetic parameter ranges	Soft-target uncertainty

4.1 Family-balanced mixture

Batches are drawn by decision family, so that no large corpus dominates by row count. The four families are *E* (evidence and classification: SNLI [Bowman et al., 2015], MNLI [Williams et al., 2018], ANLI [Nie et al., 2020a], BoolQ [Clark et al., 2019], and wide-vocabulary intent and topic sets including CLINC-150, Banking77 and HWU64 [Larson et al., 2019, Casanueva et al., 2020]); *K* (closed-book multiple-choice knowledge from nine standard sources, with TruthfulQA-MC1 [Lin et al., 2022] held out and a leakage audit against MMLU-Pro [Wang et al., 2024b, Hendrycks et al., 2020]); *W* (rubric-conditioned workflow decisions: routing, eligibility, extraction, urgency, tool selection and long policy documents); and *U* (soft-target uncertainty). The evidence corpus spans many label vocabularies at few rows per source, so the model cannot memorize a small set of label sets and has to read the rendered candidates.

The released recipe samples $E:0.40$, $K:0.15$, $W:0.35$, $U:0.10$ of every batch. The family weights are a first-order control on capability. Adding workflow data to an evidence-and-knowledge model without rebalancing costs evidence and intent accuracy, mostly through false abstention, and restoring the evidence share recovers it with workflow metrics within 1–4 points; Section 7.2 gives the sweep over the evidence weight that sets this point.

4.2 Counterfactual rubric groups

A workflow model that learns “this kind of ticket gets that label” fails the first time the policy changes. DecisionMix v2 (data_wh) is built so that this shortcut cannot fit the data: every training row belongs to a *counterfactual rubric group*, in which the same state and the same candidate set appear under at least two rubrics with distinct gold answers. Within a group the state and candidate text are identical, so any predictor that ignores the rubric assigns the same distribution to every member. For a two-member group with golds $y_1 \neq y_2$, such a predictor incurs an average cross-entropy of at least $\log 2$ nats, reached only by splitting its mass evenly between the two answers. The loss can fall below that floor only by reading the rubric.

The corpus comes from a programmatic rule engine over 12 domains crossed with 6 boolean conditions, at compositional depth levels 1 through 7, with 60,605 training rows spread evenly over levels 1–6. Level 7, which composes temporal, unit-conversion, expected-value and trade-off reasoning, is evaluation-only, and the generator never produces those compositions for training. Three whole domains, two rubric styles and one grammar per level are also held out, and the corpus is leak-checked against the public JevBench ids with 0 of 231 hits. The earlier workflow generator (data_wf) placed 26,513 of its 72,006 rows in rubric groups; DecisionMix v2 places all of them. Against a matched control never trained on it, DecisionMix v2 lifts held-out family, grammar and style accuracy from .481, .507 and .491 to .827, .881 and .888, and rubric-flip accuracy from .477 to .710 (Section 7.2).

4.3 Soft-target uncertainty corpus

Decisions with a single correct answer cannot teach a model how much probability to spread. The uncertainty corpus `data_u` (31,029 training rows, all with soft targets) supplies that signal from three sources: the validation split of UNLI [Chen et al., 2020] (2,068 rows), the ambiguous rows of AmbiEnt (401 rows, uniform target over the listed labels), and four synthetic generators with exact closed-form posteriors (28,560 rows): partial-evidence combination, a noisy-sensor Bayesian update, ordinal confusion through an inverted confusion matrix, and log-odds pooling of conflicting sources. Each generator holds out a parameter range. By type the corpus is 23,403 Noul, 7,225 Score and 401 Choice rows. ChaosNLI [Nie et al., 2020b] is used for evaluation only. The corpus trades top-1 accuracy for likelihood. In the run that added it together with DecisionMix v2, typed-decisions NLL fell from 2.06 to 1.71 and ChaosNLI NLL from 1.27 to 1.00, while a control never trained on soft targets keeps higher argmax accuracy on all three uncertainty sets (Section 7.3).

4.4 Null augmentation

A fraction of workflow rows is duplicated with the gold candidate, and any rendered catch-all option, removed; the target for the copy is $\emptyset$. The released recipe augments 20% of W rows, and results are insensitive to the fraction across $[\cdot 10, \cdot 20]$.

The fix belongs in the data because the families disagree about abstention. Under 1% of W rows carry a null target against over 20% of E rows, so a model trained on both learns two operating points. A single global temperature and null offset fitted after training cannot serve both. The same fitting procedure moves the offset in opposite directions for an evidence-only and a workflow-trained checkpoint; for the workflow-trained one it reaches +3.0, which repairs typed-decisions NLL (1.95 to 1.19) and collapses intent and topic coverage to 5–9%. Training the behavior instead lowers false abstention on TREC from .27 to .12 and raises accuracy by 2.6, 3.6, 4.3 and 6.0 points on CLINC-150, TREC, HWU64 and 20 Newsgroups (Section 7.2). The cost is soft-target NLL, since the model now places mass on $\emptyset$ where soft gold has none.

4.5 Evidence-first long states

The long-state corpus `data_wf_long` (23,318 rows of multi-paragraph policy documents, state length p10/p50/p90 of 1,190, 1,845 and 2,490 tokens) was first rendered with the case facts after the policy body. Training right-truncates states at the window, so at a 1,024-token window 98.8% of these rows lost their facts before the model saw them, and at 256 tokens all of them did. The fixed recipe regenerates the corpus facts-first, with the case inside the first 8% of the text, and drops any row longer than the window (`-drop_truncated`) so that no row is silently cut. The released v2 models train at a 2,048-token window and the 14B model at 3,072.

Placement is a controlled factor in our evaluation: Section 7.6 crosses training render with evaluation render on 605 held-out long states. The v2 releases, trained with the fixed recipe, score .947 (1.7B) and .950 (4B) on facts-first long states, against .598 and .612 for v1.

4.6 Selection on held-out likelihood

Checkpoints are selected on held-out negative log-likelihood over the uncertainty and curriculum validation sets, and a Brier term is added to the training loss. Selection on in-distribution loss favors the hard-label majority of the mixture at the expense of the soft and ordinal minority. Under accuracy-shaped selection, held-out soft-target NLL grew with backbone size, from 2.03 at 1.7B to 2.15 at 4B and 2.87 at 14B.

The 14B model also distills from its own frozen backbone read with a fixed three-shot letter rendering, $\alpha \cdot \text{CE} + \beta \cdot \text{KL}$ at temperature $T = 2$ on 63k labelled rows. The full recipe (evidence-first long states, DecisionMix v2, typed heads, likelihood-based selection with the Brier term, and distillation) takes the 14B model’s held-out Score NLL from 2.87 to 0.95 and typed-decisions NLL from 1.96 to 1.04 at the same backbone size (Section 7.3).

Table 2: Generalization axes, each with its own held-out split, and the scope a positive result licenses.

Axis	Evaluated on	A positive result licenses
Unseen examples	Held-out rows of every trained corpus	In-distribution competence
Unseen label word- ing	Label paraphrases; re-worded CLINC-150 labels	Robustness to label surface form
Unseen label space	Banking77-77, HWU64, tree-choice ($K \leq 320$), TruthfulQA-MC1	Scoring labels never trained on, within trained domains
Unseen family	3 of 15 workflow families; 3 Decision-Mix domains	Transfer across families of one generator
Unseen rubric style	2 withheld DecisionMix writing styles	Rubric conditioning beyond style memorization
Unseen grammar	1 withheld domain $\times$ level grammar	Rule execution inside the generator’s grammar
Unseen composi- tion	DecisionMix level 7 (temporal, numeric, expected value, trade-off)	Composition outside the trained grammar
External	PagerDuty, jevlogs, Mind2Web, typed-decisions, JevBench	Transfer outside our generators

5 Evaluation protocol

5.1 Evaluation sets

Training corpora are listed in Table 1. We evaluate on evidence and label-space sets (SNLI [Bowman et al., 2015], MNLI [Williams et al., 2018], ANLI [Nie et al., 2020a], BoolQ [Clark et al., 2019], CLINC-150 [Larson et al., 2019], TREC-fine, HWU64, 20 Newsgroups, and Banking77 [Casanueva et al., 2020] against all 77 labels); a fixed 1,200-item MMLU-Pro probe [Wang et al., 2024b] scored among the rendered options, with two items matched by a leakage audit excluded, and TruthfulQA-MC1 [Lin et al., 2022]; held-out splits of our generators, including the never-trained DecisionMix level 7 (880 items, uniform-guess rate .441) and ChaosNLI [Nie et al., 2020b]; and 605 held-out long states built for Section 7.6 (policy_permit, $K=2$, majority floor .612; action_select, $K=4$, floor .233). Six external sets are evaluation-only: **PagerDuty** (alert triage, yes/no, majority floor .792), **jevlogs** (log triage, yes/no, floor .697), **Mind2Web** (web-agent action choice, floor .427), **tree-choice** (taxonomy routing up to $K=320$), **typed-decisions** (400 cases across all three output types with probabilistic gold) and **JevBench** (Section 5.5). Several are class-imbalanced, so each is reported against its own majority floor. No training corpus contains any of the 231 public JevBench ids.

5.2 What “held out” means

Each held-out split excludes a whole family, domain, style, grammar or parameter range from training; none is a random validation split. Table 2 lists the axes and the scope a positive result on each licenses.

5.3 Δ_q : question dependence separated from candidate priors

A model can clear chance on a multiple-choice set by learning which option strings tend to be correct together, and accuracy and calibration do not distinguish this from answering the question.

Definition 5.1 (Question-dependence gap). For a model f and a set of decisions,

$$\Delta_q = \text{Acc}(f(x, q, A)) - \text{Acc}(f(x, \tilde{q}, A)),$$

where $\tilde{q}$ replaces the real question while the state and the candidate set A are held fixed. In the *choices-only* variant $\tilde{q}$ is an empty placeholder; in the *shuffled-question* variant Δ_q^{sh} , primary through-out, it is a real, unrelated question from the same corpus.

Δ_q extends the choices-only control of Balepur et al. [2024] to a shuffled real question, which is stricter: a degenerate placeholder can leak candidate priors through the rendering, and on one candidate-blind model the choices-only variant reads +.017 where Δ_q^{sh} reads .000. High accuracy with $\Delta_q^{\text{sh}} \approx 0$ means the model answers from the candidate set. The reference is the listwise teacher,

a fine-tuned decoder reading the options in context, with $\Delta_q^{\text{sh}} = .102$ on the 1,200 MMLU-Pro items. We treat $|\Delta_q^{\text{sh}}| < .02$ as indistinguishable from zero, since .02 is about 1.4 standard errors of an unpaired difference of proportions at $n=1,200$ and $p \approx .15$.

5.4 Metrics

We report accuracy, Brier score, expected calibration error (ECE) and negative log-likelihood (NLL) against hard or soft gold; on soft-gold sets NLL and Brier are primary. *Among- K accuracy* takes the argmax over candidates and ignores $P(\emptyset)$, and the *false-abstain rate* is the fraction of gold-present items on which $\emptyset$ wins, so abstention and discrimination errors are reported separately. *Rubric-flip* sets pair rows that share a state and candidate set under two rubrics with different gold. On MMLU-Pro counterfactual variants, *reorder Δp* is the largest change in any candidate’s probability when the options are reordered, and *pairwise $\Delta \log\text{-odds}$* the absolute shift in log-odds between the two top original candidates when three unrelated options are added; both are averaged over questions and are zero for a set-invariant scorer.

5.5 JevBench protocol

JevBench (fstandhartinger/jevbench v1.2.1) is a third-party benchmark of yes/no, choice ($K=2-6$) and ordinal decisions, each item with a state, a rubric and an exact label set. Its public release has 231 items (72 standard, 48 easy, 111 hard); 146 judge-scored items are private, and the harness ranks only submissions covering at least 95% of all items, so every JevBench number here is on the public ids and unranked. The 72 standard items are 36 states, each in two paraphrases sharing a group field, so intervals are cluster bootstraps over group (20,000 resamples): $\pm 6-13$ points on the standard tier and ± 9 on the hard tier. Comparisons between our own models use paired per-item bootstraps. Probabilities are conditioned on non- $\emptyset$: $P(\emptyset)$ is dropped and the rest renormalized. Across all models, option order follows the harness (reversing it moved one checkpoint’s standard accuracy by ± 4 points), and majority baselines are .311 standard, .284 easy and .336 hard. Leaderboard context is in Appendix A.12.

5.6 Hardware and latency

Models train on single H100 GPUs in bf16 with AdamW, effective batch 64, for 8,000–12,000 steps. Latency has two protocols. The *research harness* times one decision in process, one stream, 256-token state, state encode included and model loading excluded, for K from 2 to 256, with all sizes on one pod and one PyTorch/CUDA build. The *optimized serving path* is the released inference package, timed as warm p50 per decision against an already-cached 256-token state at $K=2$.

6 Where the decision lives

This section locates the decision inside the backbone. It compares readouts of one decision pass (Section 6.1), tests decision states computed before the candidates are known (Section 6.2), varies the layer the head reads (Section 6.3), and isolates how abstention enters the computation (Section 6.4). Unless stated otherwise, the models are 1.7B, trained for 12k steps on the evidence and knowledge mixture, and Δ_q^{sh} is measured on the 1,200-item MMLU-Pro probe against the teacher’s .102.

6.1 The contextual-candidate readout

Scoring the decision state against each candidate’s contextual state keeps the letter-logit readout’s question-conditioned signal within seed noise, with a quarter of its pairwise log-odds shift and 2–7 more points on natural language inference. Table 3 compares the three readouts of Section 3.3 on the same pass: N1 reads the next-token logits of the option letters, N2 scores h_D against an independently computed embedding of each option’s text, and N3 scores h_D against each option’s contextual state c_j^{ctx} . All arms render the candidates in the suffix, train on identical data with per-step option shuffling, and read the full 28-layer backbone; N1 and N3 each have two seeds.

The candidate’s contextual state carries the decision. N2 runs the candidates through the language model and scores the same h_D with the same scorer; it differs from N3 only in scoring against

Table 3: Three readouts of one decision pass, full-depth 1.7B backbone, identical data and steps; N1 and N3 at two seeds (seed=0/seed=1), N2 at one. Teacher: a fine-tuned decoder reading the options in context. Order measures are zero for a set-invariant scorer.

	N1 s0	N1 s1	N2	N3 s0	N3 s1	Teacher
Δ_q^{sh} (primary)	.114	.104	.058	.117	.094	.102
MMLU-Pro among- K	.325	.318	.198	.314	.301	.310
Banking77, $K=77$	.531	–	.012	.497	–	–
SNLI	.838	.831	.854	.863	.851	–
MNLI	.684	.664	.742	.752	.719	–
ANLI	.377	.393	.410	.431	.437	–
BoolQ	.769	.764	.775	.737	.746	–
Reorder Δp	.165	.165	.059	.101	.109	.097
Pairwise $\Delta\log\text{-odds}$	.463	.443	.379	.126	.094	.179

Table 4: Candidate-blind variants: six factorizations and objectives, two tap depths. “Shuffled” replaces the real question with an unrelated one, candidate set fixed; Δ_q^{sh} is the difference. $n=1,200$ MMLU-Pro items.

Variant	Factorization / objective	among- K	shuffled	Δ_q^{sh}
z1	single decision vector, bilinear scorer	.171	.147	+.023
zr	8 probes, token-level max-similarity	.157	.182	–.026
zr_set	+ $O(K)$ cross-attention over candidate tokens	.147	.151	–.003
ms_ctrl	single-set distillation, steps-matched control	.124	.147	–.023
ms_zr_set	same- Z multi-set, $\Delta\log\text{-odds}$ targets	.130	.130	+.000
zr (full depth)	8 probes, tap at layer 28 of 28	.147	.156	–.008
Teacher	listwise decoder, options in context	.310	.208	+.102

an independent embedding of the option text. It loses 13 points of among- K accuracy against N1 (.198 vs. .325), half of the question-conditioned signal (Δ_q^{sh} .058 vs. .114), and collapses on Banking77 at $K=77$ (.012 against .531 and .497 for N1 and N3). The pattern fits a decision state that encodes which rendered slot holds the answer: the letter embedding and the contextual state both carry slot identity, and an independently embedded label carries only its meaning.

N3 matches N1 on question dependence and is far less sensitive to the candidate set. Across seeds N1 averages $\Delta_q^{\text{sh}} = .109$ and N3 .106, against the teacher’s .102; the seed-to-seed spread is .01–.02 for both readouts, so the two are within seed noise of each other and of the teacher. Among- K accuracy is 1–2 points lower for N3 (.314/.301 vs. .325/.318). Adding three unrelated options shifts N3’s pairwise log-odds by .09–.13 against .44–.46 for N1, a quarter of the shift, and reordering moves N3’s probabilities by .10–.11 against .165. On evidence tasks N3 leads in both seeds by 2 points on SNLI, 6–7 on MNLI and 4–5 on ANLI, and trails by 2–3 on BoolQ. Per-step option shuffling is shared by both arms, so the difference in order sensitivity is the readout’s. N3 is the readout Typical deploys; its Choice output is order-robust, and exact order invariance is the Bernoulli head’s contract (Section 7.1).

6.2 Candidate-blind decision states

In every configuration we tested, question-conditioned capability appeared only when the candidates were processed inside the forward pass. Six candidate-blind variants, spanning three head factorizations, two distillation objectives and two tap depths, all have Δ_q^{sh} within .03 of zero, while the letter-logit and contextual readouts and the teacher, which compute over the candidates in context, sit at .09–.12 and even the semantic readout N2 keeps .058 (Figure 2). A candidate-blind variant computes a decision state $Z(x, q)$ before the candidates are seen and scores each candidate against it with a learned head; each is distilled from the listwise teacher.

The result holds across every lever we varied (Table 4). A single decision vector retains as much as eight probes. Cross-attention from the probes over all candidate tokens leaves Δ_q^{sh} at –.003 while improving null quality (CLINC out-of-scope accuracy .568 $\rightarrow$.727). Supervising the change in the teacher’s log-odds across seven candidate sets for the same question gives Δ_q^{sh} of exactly

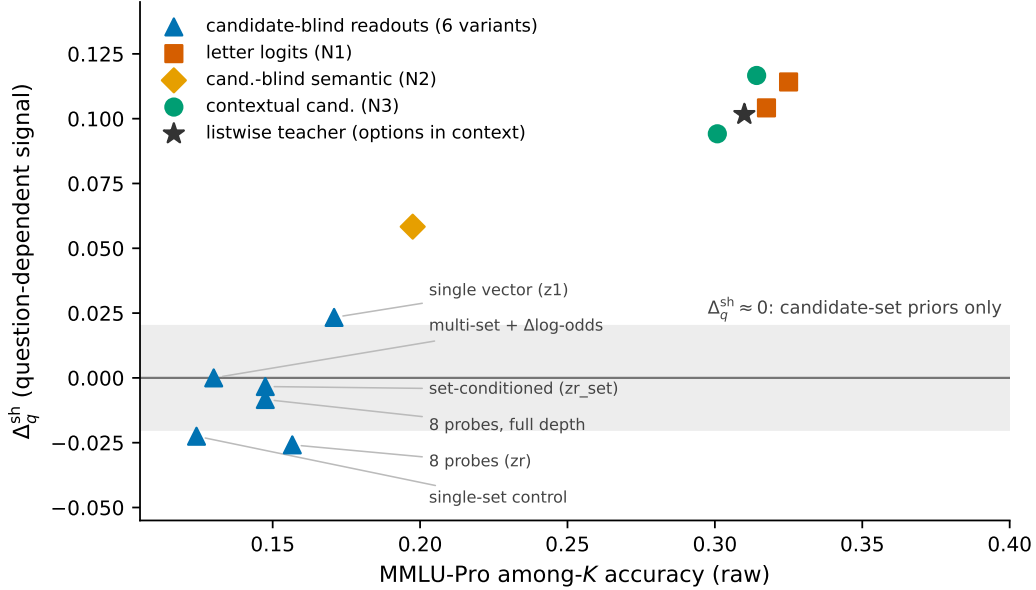


Figure 2: Raw MMLU-Pro among- K accuracy against Δ_q^{sh} , the accuracy lost when the real question is replaced by a shuffled one with the state and candidate set fixed ($n=1,200$; shaded: the $\pm .02$ band of Section 5.3). The six candidate-blind variants cluster at zero at every head capacity, objective and depth. The letter-logit (N1) and contextual (N3) readouts and the listwise teacher carry .09–.12; the semantic readout N2 sits between. Accuracy alone does not separate the groups.

zero, against $-.023$ for its steps-matched single-set control. Reading the full 28 layers instead of layer 20 leaves Δ_q^{sh} at $-.008$, lowers among- K accuracy to .147, and regresses the evidence tasks (SNLI/MNLI/ANLI/BoolQ .881/.807/.462/.712 against .905/.873/.541/.826 at layer 20; Table 8). Two variants fall just outside the $\pm .02$ band, in opposite directions ($z1 +.023$, $zr -.026$), which is the spread 1,200 items allow; every variant is separated from the teacher’s .102. The one mechanistic alternative the data leave open is an optimization failure in the set-conditioning path, whose zero-initialized cross-attention stayed inert through 4k steps of direct supervision.

What a candidate-blind state does learn explains why accuracy hides the gap. The variants inherit the teacher’s candidate priors and its calibration. Their accuracy of .12–.17 sits above the .10 chance rate of ten-option questions and comes from which option strings tend to be correct together; ECE on the knowledge probe falls from .39 for an earlier baseline to .04; and they are exactly set-stable (reorder Δp and pairwise $\Delta \log\text{-odds}$ both zero) where the teacher is not. On accuracy and calibration alone the factorization looks sound. Δ_q^{sh} shows that the question-dependent component lives in candidate-contextual computation, which sets the design principle for Typical: render the candidates inside the backbone’s forward pass. The candidate-blind state keeps one role, shortlisting large candidate sets, a task that runs on priors (Section 7.5).

6.3 Mid-depth readout

Reading at 71% of depth gives one model that carries both evidence reasoning and question-conditioned knowledge. The first full-scale model in this project read the backbone’s last layer and scored 58.6 on SNLI, the hypothesis-only baseline: it predicted entailment without using the premise. Moving the tap to layer 20 of 28 lifted it to 65.3 frozen and 68.2 with LoRA. The same effect appears in the contextual readout under a controlled comparison. With head, data and steps fixed, reading N3 at layer 20 instead of 28 raises SNLI/MNLI/ANLI/BoolQ from .863/.752/.431/.737 to .906/.865/.519/.834, CLINC-150 from .740 to .862, and lowers validation NLL from .430 to .341, while Δ_q^{sh} stays at .118 against .117 at full depth. Evidence accuracy lands within 1–3 points of a candidate-blind energy-scoring baseline on every evidence set, and 8 points above it on CLINC-150, so the evidence and knowledge regimes that had needed separate models are served by one. For a candidate-blind state the same change in depth moves nothing (Section 6.2).

The final layers of a causal language model specialize for next-token prediction, and that the most useful features for a downstream decision sit below them agrees with work on layer pruning and probing [Gromov et al., 2024, Men et al., 2025, Tenney et al., 2019, Belrose et al., 2023]. For a deployed decision interface the choice is decisive and interacts with the readout: depth recovers evidence for the candidate-aware readout and leaves a candidate-blind state unchanged. Every checkpoint we report uses the 71% tap, ported across sizes as layer 20 of 28, 26 of 36 and 28 of 40.

6.4 Abstention as a gate

Abstention must be a gate and never a rendered candidate. A rendered “none of the above” line competes for softmax mass with the real candidates, so its share depends on how many there are. We isolate this with a matched pair: the same model (N3 at layer 20, with the factored gate of Equation (2)) with and without a rendered $\emptyset$ line in the suffix. With the line, the abstention behavior is a function of K : on CLINC-150 with the gold label absent the model abstains with probability .02 at $K=5$ and 1.00 at $K=150$, where it abstains on every item, and on Banking77 at $K=77$ null AUROC is .00 by the same mechanism.

Removing the line removes both degenerate endpoints and makes the K -sweep smooth: null AUROC on CLINC-150 is .98/.95/.92/.95 at $K = 5/20/50/150$ and on Banking77 .90/.83/.69 at $K = 5/20/77$. Downstream, MMLU-Pro false abstention falls from .692 to .003 and accuracy rises from .133 to .353, the KL divergence to the teacher’s distribution on MMLU-Pro falls from 2.43 to .27, Banking77 with all 77 labels rises from .063 to .579, and CLINC out-of-scope accuracy from .639 to .798, with Δ_q^{sh} unchanged (.118 vs. .122) and evidence tasks at the same level (Table 9). The false-abstain change is far outside its seed-to-seed variation of .05–.10. The price is 2–7 points on intent and topic label spaces (HWU64 .801 $\rightarrow$.757, 20NG .589 $\rightarrow$.515) and a rise in reorder Δp from .077 to .123. The gate’s algebraic properties hold by construction (Section 3.6); this pair shows their empirical consequence, and null augmentation supplies the training signal that teaches the gate when to fire (Section 7.2). This configuration, contextual readout at 71% depth with a gate and no rendered $\emptyset$ line, is the architecture of every checkpoint in Section 7.

7 Results

This section reports what the typed contracts, the training recipe and scale deliver, what a decision costs, and two controlled experiments on how inputs are rendered. All checkpoints share the architecture fixed in Section 6.

7.1 Typed output contracts

Each output type delivers the property its contract names, with backbone, tap and readout fixed. We compare matched fine-tunes from one parent checkpoint, each trained for 3k steps on the rows of its own type, against a control that treats the same rows as plain Choice; full tables are in Appendix A.5.

Score. The ordinal-smoothed target ($\tau = 0.7$) adds no parameters and improves every probability-quality measure on held-out urgency at flat accuracy (.505 $\rightarrow$.508): NLL 2.07 $\rightarrow$ 1.23, Brier .79 $\rightarrow$.68, ECE .35 $\rightarrow$.19, ordinal MAE .58 $\rightarrow$.55. On JevBench’s 12 ordinal items the two models make identical per-item predictions, so the gain is in how mass is spread over adjacent levels. On an external severity set it also lifts accuracy from .853 to .939 and cuts MAE from .15 to .06.

Noul. The Bernoulli head renders no candidates, so label order cannot enter its computation. Reversing the labels of a two-way Choice over {no, yes} moves $P(\text{yes})$ by up to .55 (mean .01–.15 across sets); the Bernoulli head moves it by exactly zero on all nine test sets. It is better or equal on six of seven yes/no sets and clears the untouched PagerDuty floor by a wide margin: accuracy .602 $\rightarrow$.886 against a .792 majority floor, NLL 1.04 $\rightarrow$.28. Only rows whose candidate set is exactly {yes, no} take the Bernoulli path.

7.2 The training distribution

Decision-type balance, counterfactual rubric groups and null targets each move a separate axis. All comparisons here are matched 1.7B runs that differ only in data or sampler.

Family balance. Adding workflow data at family weights E .35 / K .25 / W .40 cost 11 points on CLINC-150 and 10 on TREC-fine, almost all of it false abstention (CLINC-150 .05 $\rightarrow$.17). In a sweep of the evidence share (Table 10), $E \geq .45$ brings NLI and BoolQ back within 1 point of the evidence-only base and intent and topic sets within 3–4, with workflow metrics unchanged within 1–4.

Counterfactual rubric groups. DecisionMix v2 places every training row in a group that shares a state and candidate set across rubrics with different gold answers (Section 4.2). Added together with the uncertainty corpus, against a control never trained on either, it lifts held-out families, grammars and rubric styles from .481/.507/.491 to .827/.881/.888, and accuracy on held-out rubric-flip pairs from .477 to .710. Accuracy under a shuffled rubric falls from .446 to .405: the model now depends on the rubric it is given. Level-7 composition, which the generator never produces, moves from .477 to .498 (Section 9). The original workflow sets give up 3–5 points at a fixed workflow share now split over four corpora.

Null augmentation. Duplicating 20% of workflow rows with the gold candidate removed and target $\emptyset$ lowers false abstention on CLINC-150 from .10 to .08 and on TREC-fine from .27 to .12, and raises accuracy by 2.6, 3.6, 4.3 and 6.0 points on CLINC-150, TREC-fine, HWU64 and 20NG. MMLU-Pro among- K returns to the evidence-only base (.353), with the highest Δ_q^{sh} of any mixture (.143). The model now spends mass on $\emptyset$ where soft gold has none, so held-out Noul drops 4 points and soft-target NLL rises. Any augmentation fraction in [.10, .20] gives the same model within noise.

Soft targets. The uncertainty corpus, added in the same run, trades top-1 accuracy for likelihood (ChaosNLI NLL 1.27 $\rightarrow$ 1.00 with accuracy .584 $\rightarrow$.537), and typed-decisions NLL improves from 2.06 to 1.71.

7.3 Probability quality

Probability quality is indexed by decision type, so it is set by training and checkpoint selection.

One global temperature and null-logit offset, fit jointly on held-out soft-target rows, wants opposite corrections for two checkpoints. For an evidence-and-knowledge checkpoint the fitted offset is -1.75 , which cuts CLINC-150 false abstention from .053 to .022. For a workflow-trained checkpoint it is $+3.0$, which repairs typed-decisions NLL (1.95 $\rightarrow$ 1.19) and held-out Score NLL (2.03 $\rightarrow$ 1.09) and collapses intent and topic coverage to 5–9%. Null augmentation and per-type targets supply the correction at the level of data and objective instead (Sections 7.1 and 7.2).

Selection criterion. Under accuracy-shaped checkpoint selection, held-out Score NLL grew with scale: 2.03 at 1.7B, 2.15 at 4B and 2.87 at 14B. The 14B recipe bundle (facts-first long states with no truncation, DecisionMix v2 and the uncertainty corpus, typed heads, a Brier term with selection on held-out uncertainty and curriculum NLL, and frozen-teacher distillation) takes the same 14B backbone to held-out Score NLL 0.95 and typed-decisions NLL 1.04 (from 1.96), with JevBench hard-tier Brier .85 $\rightarrow$.66 (Figure 3). The matched arm without distillation reaches the same held-out Score NLL level, and the 1.7B and 4B checkpoints trained with typed heads and DecisionMix sit at 0.97–1.03.

7.4 Scaling

Table 5 and Figure 4 run one recipe at four Qwen3 sizes [Yang et al., 2025a] and on Qwen3.5-4B, each beside a frozen control: the same base model, untrained, reading next-token letter logits over the rendered options with three exemplars.

Training beats the frozen backbone on the standard tier at every Qwen3 rung, by 16.6, 2.8, 11.1 and 11.2 points at 1.7B, 4B, 8B and 14B. The sign is consistent across all four rungs; each gap on its own sits inside its interval, and the paired test at 14B gives frozen-minus-trained $-.111$ $[-.250, +.014]$, $p = .10$. The 14B checkpoint reaches .931 on the standard tier. Question-conditioned knowledge grows with the backbone: Δ_q^{sh} rises from .127 at 1.7B to .193 at 4B and .235 at 14B, 2.3 times the 1.7B teacher’s. The second-generation backbone is the strongest point per parameter: Qwen3.5-4B, released as typical-medium v2, scores .861 standard and .495 hard, above every Qwen3 checkpoint we trained on the hard tier, and 9.7 standard-tier points above its own frozen control. The Qwen3-8B rung underperforms its neighbours in both frozen and trained

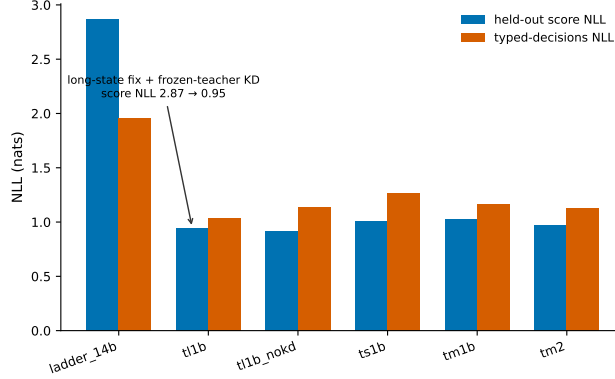


Figure 3: Held-out Score and typed-decisions NLL (held-out sets, not JevBench). ladder_14b is the 14B under the earlier recipe; t11b is the same backbone under the recipe bundle of Section 7.3, and t11b_nokd drops distillation from that bundle. ts1b, tm1b and tm2 are the 1.7B and 4B checkpoints with typed heads and DecisionMix.

Table 5: Scaling ladder with frozen controls. JevBench: public 231-id subset, unranked; 95% intervals are cluster bootstraps over the paraphrase group (Section 5.5). MMLU- K : MMLU-Pro among- K . NLL: held-out Score. ms: research-harness single decision at $K=2$, one pod and build. Frozen: no training, letter logits, 3 exemplars. Small and medium are the typical-small and typical-medium releases. Checkpoints: ts1b, ts1c, tm1b, ladder_8b, t11b, tm2.

Model	JevBench std [95% CI]	JevBench hard [95% CI]	MMLU- K	Δ_q^{sh}	NLL	ms
Qwen3-1.7B (small v1)	.694 [.569, .819]	.432 [.342, .523]	.343	.127	1.01	45
small v2	.708 [.569, .833]	.432 [.342, .523]	.338	.127	1.08	—
frozen, 3-shot	.528 [.389, .667]	.369 [.279, .459]	—	—	—	—
Qwen3-4B (medium v1)	.806 [.694, .903]	.423 [.333, .514]	.458	.193	1.03	57
frozen, 3-shot	.778 [.667, .889]	.441 [.351, .532]	—	—	—	—
Qwen3-8B	.667 [.528, .792]	.396 [.306, .486]	.302	.093	1.67	70
frozen, 3-shot	.556 [.417, .694]	.369 [.279, .459]	—	—	—	—
Qwen3-14B	.931 [.861, .986]	.450 [.360, .541]	.498	.235	0.95	59
frozen, 3-shot	.819 [.708, .917]	.559 [.468, .649]	—	—	—	—
Qwen3.5-4B (medium v2)	.861 [.778, .944]	.495 [.405, .586]	.429	.194	0.97	—
frozen, 3-shot	.764 [.639, .875]	.495 [.405, .586]	—	—	—	—

form under the identical recipe, which marks it as a property of that checkpoint. The hard tier, where the frozen 14B leads, is discussed in Section 9.

7.5 Cost

Decision latency is nearly flat in backbone size. On the research harness a single decision at $K=2$ costs 45, 57 and 59 ms at 1.7B, 4B and 14B: an $8\times$ larger backbone is $1.3\times$ slower, because the cached state and a short suffix dominate. The cost stays flat to $K=32$ (46, 58 and 62 ms) and reaches 106 ms at 1.7B and 156 ms at 14B at $K=256$. For M questions against one cached state the cost is $O(|x|) + M \cdot O(|q| + |A|)$, so the batched marginal cost per question grows with size and candidate tokens: 2.7 and 3.9 ms at $K=2$ for 1.7B and 14B, 28.0 and 109.7 ms at $K=256$.

On the optimized serving path a warm decision takes 15.5–17 ms at 1.7B. The released inference package originally deep-copied the prefix key-value cache on every decision. A zero-copy strided view, with bit-identical output, plus cached masks and position tensors brings warm p50 from 21–25 to 15.5–17 ms at 1.7B and from 26–27 to 19–21 ms for the Qwen3-4B model (Figure 7); the Qwen3.5-4B model measures 34–46 ms with reference DeltaNet kernels.

Large candidate sets go through a shortlist. With 256 questions against one state, the contextual readout’s per-question cost grows from 1.07 ms at $K=2$ to 43.8 ms at $K=256$, while a candidate-blind scorer stays at 0.97–1.06 ms. Sets in the hundreds are served by that scorer producing a

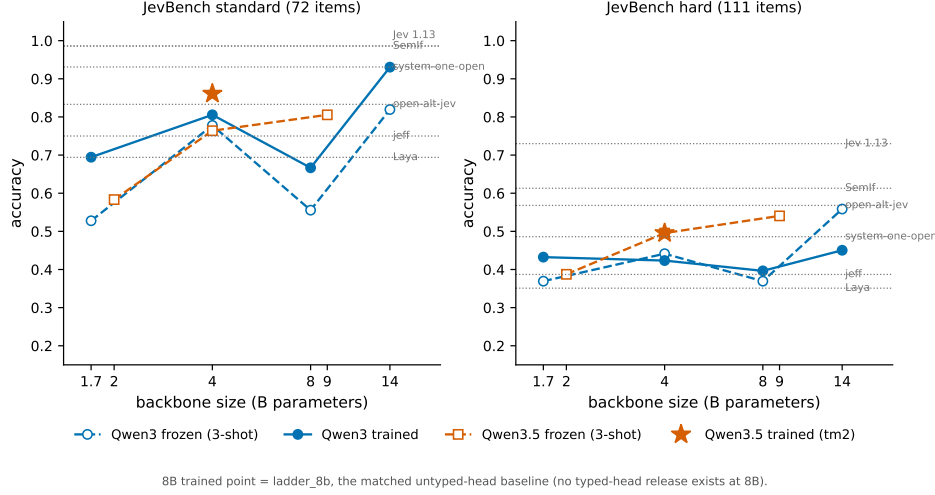


Figure 4: JevBench standard and hard accuracy against backbone size, public 231-id subset, un-ranked. Qwen3 trained (solid) and frozen with three exemplars (dashed); Qwen3.5 frozen controls at 2B/4B/9B and the trained Qwen3.5-4B (star). Dotted lines mark other public entries on the same ids, whose rendering and protocols differ (Appendix A.12). Intervals are in Table 5.

Table 6: Training render $\times$ evaluation render on 605 held-out long states, no truncation at evaluation. Diagonal cells are matched; the matched-vs-matched comparison isolates the training effect. Majority floors: policy_permit .612 ($K=2$, $n=330$), action_select .233 ($K=4$, $n=275$). 14B arms: t11b_nokd (facts-first) and ladder_14b (facts-last).

Trained on	Scored on	overall	policy_permit	action_select
<i>1.7B, single-variable arms</i>				
facts-first	facts-first (<i>matched</i>)	.942	.933	.953
facts-first	facts-last	.797	.788	.807
facts-last	facts-first	.640	.615	.669
facts-last	facts-last (<i>matched</i>)	.830	.842	.815
<i>14B corroboration</i>				
facts-first	facts-first (<i>matched</i>)	.997	.997	.996
facts-first	facts-last	.921	.933	.905
facts-last	facts-first	.851	.839	.865
facts-last	facts-last (<i>matched</i>)	.919	.930	.905

bounded shortlist for the contextual readout to decide, the role left to the candidate-blind state of Section 6.2, since shortlisting runs on priors.

7.6 Evidence placement in long states

Training with the evidence placed first adds 11.2 points on held-out long states, isolated by a render-crossed design and replicated at a second seed. The original long-state workflow corpus rendered the case facts after the policy text, and the loader right-truncated states to the training window: at the 1,024-token window used for the first releases, 98.8% of long-policy rows lost their facts (Section 4.5). Two 1.7B arms with byte-identical flags train on the same 23,318 rows, rendered facts-first or facts-last, with truncation active as at release. Both are scored on 605 held-out long states under both renders with no truncation at evaluation (4,096-token window; state p50 1,965 tokens, max 2,843). Matched cells measure what training cost; off-diagonal cells measure render mismatch (Table 6).

At 1.7B the matched-condition gap is $+.112$ [$+.077, +.147$], $p = 5 \times 10^{-10}$ (policy_permit $+.091$ [$+.043, +.139$], action_select $+.138$ [$+.086, +.190$]), and a second seed of the whole pair gives $+.111$. At 14B the gap is $+.078$ [$+.055, +.100$], $p = 7 \times 10^{-12}$, smaller because the facts-first

Table 7: Rendering alone, frozen weights. Identical Qwen3.5 base checkpoints and items (JevBench, public 231-id subset, unranked), letter-logit readout in both columns; only the state and option rendering differs. Δ : paired cluster bootstrap over the per-item difference, 20,000 resamples.

Tier	Backbone	our rendering	replicated rendering	paired Δ [95% CI], p
standard	Qwen3.5-2B	.583	.597	+ .014 [−.194, +.208], $p=.95$
	Qwen3.5-4B	.764	.847	+ .083 [+ .014, +.167], $p=.048$
	Qwen3.5-9B	.806	.931	+ .125 [+ .056, +.208], $p<.001$
hard	Qwen3.5-2B	.387	.441	+ .054 [−.045, +.153], $p=.33$
	Qwen3.5-4B	.495	.468	− .027 [−.117, +.063], $p=.64$
	Qwen3.5-9B	.541	.595	+ .054 [−.027, +.135], $p=.22$

model is at ceiling (.997). The two 14B checkpoints differ in the rest of the recipe bundle as well, so the 14B pair corroborates the direction and size at scale and the 1.7B pair is the single-variable test.

The fixed checkpoints are the current releases. With the case facts before the policy body, the first releases scored .598 (1.7B) and .612 (4B), and on the yes/no family they sat at or below its .612 majority floor (.536 and .555). The v2 releases, trained on the fixed corpus, score .947 and .950 (policy_permit .948 and .967), and hold their facts-last accuracy (.790 vs. .798 at 1.7B, .879 vs. .866 at 4B). JevBench does not register the change: typical-small v2 and v1 read .708 and .694 on its standard tier (Table 5). The benchmark’s 19-item long_policy cell could not resolve the effect either, returning $p = .232$ at one seed and a tie at the other (Appendix A.9).

7.7 Rendering as an experimental factor

Changing only the rendering moves a frozen 9B model 12.5 points. Scoring identical public JevBench items with identical frozen Qwen3.5 checkpoints and the same letter-logit readout, changing only the layout of the state and options to a replication of another published system’s format, lifts the standard tier by 1.4, 8.3 and 12.5 points at 2B, 4B and 9B (Table 7). The effect survives a paired per-item test at 4B ($p = .048$) and 9B ($p < .001$), and no hard-tier change separates at any size.

Inside one trained model, render mismatch costs more than the data defect. In the off-diagonal cells of Table 6, where the evidence is fully visible and only its position changes, switching the evaluation render against the training render costs 14.5 points for the facts-first 1.7B model and 19.0 for the facts-last one, which lands on its majority floor on policy_permit (.615 against .612). At 14B the same switch costs 7.6 and 6.8 points, so capacity buys render tolerance that the 1.7B pair alone would not reveal. Our own checkpoints give the sharpest illustration: the earlier-recipe 14B scores .053 on JevBench’s 19-item long_policy family and .919 on the 605 held-out long states in its own training render. The pair also changes the item set, and the gap compounds truncation damage, render mismatch against the benchmark’s fixed format and a 19-item sample.

A benchmark score therefore measures capability jointly with interface compatibility, and we read our own numbers that way. The frozen controls of Table 5 use a letter-logit rendering we did not sweep and the trained models the native one, so frozen-versus-trained comparisons carry a rendering factor of the size shown here; leaderboard context sits in Appendix A.12 for the same reason.

8 Discussion

Decisions live in candidate-contextual states at mid-depth. Three measurements point at the same place. Question-conditioned capability appears only when the candidates are processed inside the language model’s own forward pass (Section 6.2); among readouts that do process them, the candidate’s contextual state carries the decision and an independently embedded copy of the same text does not (Section 6.1); and the layer that holds it sits before the top of the stack specializes for next-token prediction (Section 6.3). This refines the cross-encoder versus dual-encoder picture. In every configuration we tested, a candidate-independent state reproduced the teacher’s candidate priors and calibration, and the question-dependent component appeared only with the candidates in context. The reusable-state formulation remains valuable for shortlisting large candidate sets, a sub-task that runs on priors (Section 7.5), and Δ_q is the measurement that tells the two roles

apart. We recommend reporting it for any architecture that scores candidates against a precomputed representation.

An output type is a contract, and structure enforces it. Each typed primitive fixes a property that post-hoc calibration cannot reach. The Bernoulli head is order-invariant because there is no candidate order to be sensitive to; the ordinal target moves probability mass toward adjacent levels without changing a decision; the gate keeps null AUROC between .92 and .98 from $K = 5$ to $K = 150$, where a rendered abstain option collapsed at both ends. The same lesson appears in reverse at the output layer: one global temperature and null offset, fit on held-out soft targets, wants opposite corrections for an evidence-trained and a workflow-trained checkpoint (Section 7.3). Probability quality for a decision model is indexed by decision type, and the productive place to shape it is the head, the target and the training data.

The distribution of decision types is a first-order lever. Several of the largest effects in this paper come from data and selection with the architecture held fixed: counterfactual rubric groups (35–40 points on held-out families, grammars and styles), null augmentation (2.6–6.0 points on label-space accuracy, from teaching the model when to abstain), family-balanced sampling (bringing evidence and intent accuracy back to within 1 and 3–4 points of the evidence-only base after an unbalanced mix cost 10–11), evidence-first rendering of long states (11.2 points), and a 14B recipe built around calibration-based checkpoint selection (a threefold reduction in held-out ordinal NLL). Decision models are trained on a mixture of decision *shapes*, and each shape the mixture contains is one the model learns.

A benchmark score measures capability together with interface compatibility. Changing only the rendering moves a frozen 9B model 12.5 points on the same items (Section 7.7), more than the gap between many systems a benchmark is used to rank. Two practices made our own measurements decisive. The first is to cross training render with evaluation render, so that data effects and compatibility effects separate. The second is to build an evaluation set with the power to resolve the effect in question: 605 generated, leak-checked long states settled at $p < 10^{-9}$ a question that a 19-item benchmark cell returned as $p = .232$. Pre-registration protects a decision rule only when the metric it names can resolve the effect it gates.

What a single pass computes. Typical decides in one forward pass of fixed depth, with no intermediate tokens. Fixed-depth transformers correspond to constant-depth threshold circuits [Merrill and Sabharwal, 2023a], while conditioning on self-generated intermediate tokens extends what they can compute [Merrill and Sabharwal, 2023b, Feng et al., 2023]. The decisions that remain hard for every model we trained, temporal arithmetic, unit conversion, expected value and multi-attribute trade-offs (Appendix A.9), have the serial shape that picture singles out. That makes a concrete next step: keep the typed readout and the cached state, and add a bounded amount of intermediate computation before the decision token, in the form of extra passes or a short scratchpad. The interface in this paper is the natural terminal layer for such a system, since a decision still has to end in a typed, thresholdable distribution over the options the caller supplied.

9 Limitations

Scope. Typical makes text-only, bounded decisions over candidate sets that fit in a short suffix: categorical, ordinal and Bernoulli contracts with abstention. It does not plan, search, call tools or read images. Per-query cost grows with the rendered candidate tokens, so candidate sets in the hundreds are served through a shortlisting front end (Section 7.5).

Multi-step composition. Decisions that require carrying an intermediate value through several steps are the open frontier. On DecisionMix level 7 (temporal, numeric, expected-value and trade-off composition, never trained) the v1 1.7B and 4B models score .493 and .544 against a .441 guess rate, and the 14B model .620. On the JevBench hard tier a frozen 14B backbone read with three in-context exemplars scores above our trained 14B checkpoint (paired difference $+.108$ [$+.027, +.189$], $p = .015$), while the trained model leads on the standard tier. Appendix A.9 gives the per-family breakdown. Section 8 describes the extension this points to.

Statistics. The readout comparison in Section 6.1 and the evidence-first result in Section 7.6 are replicated across seeds; other rows are single training runs. JevBench numbers use the 231 public items and are not a ranked leaderboard entry. Its intervals are ± 6 –13 points on the standard tier (36 independent states) and ± 9 points on the hard tier (111 items), so we base comparisons between our own checkpoints on paired tests and on larger purpose-built sets.

Generalization and invariance. “Runtime-defined labels” refers to the axes of Table 2. The rubric-generalization results share one rule engine between training and held-out families. Choice is order-robust rather than order-invariant: the contextual readout reduces order sensitivity relative to the letter interface, per-step option shuffling during training contributes to that robustness, and only Noul is exactly invariant. The pairwise-odds property of Equation (2) holds among non-null candidates; the abstention gate is set-dependent by design.

Ladder and release. The size ladder is one recipe on one backbone family (Qwen3) with a Qwen3.5-4B checkpoint as a second-generation point; the Qwen3-8B rung underperforms its neighbours in both frozen and trained form. Inference code, weights and evaluation outputs are public; the training code is not yet released.

10 Conclusion

Typical reads decisions directly out of a pretrained causal language model. A state is encoded once, a question and its runtime candidates are read in a short suffix, and a head at mid-depth returns a typed distribution with a factored abstention event, at 45 ms per decision at 1.7B and 59 ms at 14B. The design rests on measured answers to where the decision lives: inside the forward pass over the candidates, in their contextual states, below the layers that specialize for generation. The training recipe shows that decision types, counterfactual rubrics, abstention targets and selection criteria shape a decision model as strongly as its architecture, and the render-crossed evaluation shows how to measure such effects cleanly. The models and the inference package are public, and the same interface is the natural terminal layer for decision systems that add intermediate computation before the typed readout.

References

- Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes, 2016. URL <https://arxiv.org/abs/1610.01644>.
- Amanda Askell, Yuntao Bai, Anna Chen, Dawn Drain, Deep Ganguli, Tom Henighan, Andy Jones, Nicholas Joseph, Ben Mann, Nova DasSarma, Nelson Elhage, Zac Hatfield-Dodds, Danny Hernandez, Jackson Kernion, Kamal Ndousse, Catherine Olsson, Dario Amodei, Tom Brown, Jack Clark, Sam McCandlish, Chris Olah, and Jared Kaplan. A general language assistant as a laboratory for alignment, 2021. URL <https://arxiv.org/abs/2112.00861>.
- Nishant Balepur, Abhilasha Ravichander, and Rachel Rudinger. Artifacts or abduction: How do llms answer multiple-choice questions without the question? In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 1030810330. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.555. URL <http://dx.doi.org/10.18653/v1/2024.acl-long.555>.
- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207219, 2022. ISSN 1530-9312. doi: 10.1162/coli_a_00422. URL http://dx.doi.org/10.1162/coli_a_00422.
- Nora Belrose, Igor Ostrovsky, Lev McKinney, Zach Furman, Logan Smith, Danny Halawi, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens, 2023. URL <https://arxiv.org/abs/2303.08112>.
- Divyanshu Bhatt, Abhinav Joshi, Ujjaval Patel, and Ashutosh Modi. From representation to choice: Tracing decision emergence across languages in LLMs. In Maria Liakata, Viviane P. Moreira, Jiajun Zhang, and David Jurgens, editors, *Findings of the Association for Computational Linguistics: ACL 2026*, pages 42423–42442, San Diego, California, United States, July 2026. Association for

- Computational Linguistics. ISBN 979-8-89176-395-1. doi: 10.18653/v1/2026.findings-acl.2105. URL <https://aclanthology.org/2026.findings-acl.2105/>.
- Dan Biderman, Jacob Portes, Jose Javier Gonzalez Ortiz, Mansheej Paul, Philip Greengard, Connor Jennings, Daniel King, Sam Havens, Vitaliy Chiley, Jonathan Frankle, Cody Blakeney, and John P. Cunningham. Lora learns less and forgets less, 2024. URL <https://arxiv.org/abs/2405.09673>.
- Samuel R. Bowman, Gabor Angeli, Christopher Potts, and Christopher D. Manning. A large annotated corpus for learning natural language inference. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, page 632642. Association for Computational Linguistics, 2015. doi: 10.18653/v1/d15-1075. URL <http://dx.doi.org/10.18653/v1/d15-1075>.
- GLENN W. BRIER. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):13, January 1950. ISSN 1520-0493. doi: 10.1175/1520-0493(1950)078<0001:vofeit>2.0.co;2. URL [http://dx.doi.org/10.1175/1520-0493\(1950\)078<0001:vofeit>2.0.co;2](http://dx.doi.org/10.1175/1520-0493(1950)078<0001:vofeit>2.0.co;2).
- Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwu Peng, Jason D. Lee, Deming Chen, and Tri Dao. Medusa: Simple llm inference acceleration framework with multiple decoding heads, 2024. URL <https://arxiv.org/abs/2401.10774>.
- Qingqing Cao, Harsh Trivedi, Aruna Balasubramanian, and Niranjan Balasubramanian. Deformer: Decomposing pre-trained transformers for faster question answering. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, page 44874497. Association for Computational Linguistics, 2020a. doi: 10.18653/v1/2020.acl-main.411. URL <http://dx.doi.org/10.18653/v1/2020.acl-main.411>.
- Wenzhi Cao, Vahid Mirjalili, and Sebastian Raschka. Rank consistent ordinal regression for neural networks with application to age estimation. *Pattern Recognition Letters*, 140:325331, December 2020b. ISSN 0167-8655. doi: 10.1016/j.patrec.2020.11.008. URL <http://dx.doi.org/10.1016/j.patrec.2020.11.008>.
- Iñigo Casanueva, Tadas Teminas, Daniela Gerz, Matthew Henderson, and Ivan Vuli. Efficient intent detection with dual sentence encoders. In *Proceedings of the 2nd Workshop on Natural Language Processing for Conversational AI*, page 3845. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.nlp4convai-1.5. URL <http://dx.doi.org/10.18653/v1/2020.nlp4convai-1.5>.
- Charlie Chen, Sebastian Borgeaud, Geoffrey Irving, Jean-Baptiste Lespiau, Laurent Sifre, and John Jumper. Accelerating large language model decoding with speculative sampling, 2023. URL <https://arxiv.org/abs/2302.01318>.
- Tongfei Chen, Zhengping Jiang, Adam Poliak, Keisuke Sakaguchi, and Benjamin Van Durme. Uncertain natural language inference. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, page 87728779. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.acl-main.774. URL <http://dx.doi.org/10.18653/v1/2020.acl-main.774>.
- Zhoujun Cheng, Jungo Kasai, and Tao Yu. Batch prompting: Efficient inference with large language model apis. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: Industry Track*, page 792810. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.emnlp-industry.74. URL <http://dx.doi.org/10.18653/v1/2023.emnlp-industry.74>.
- C. Chow. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1):4146, January 1970. ISSN 1557-9654. doi: 10.1109/tit.1970.1054406. URL <http://dx.doi.org/10.1109/tit.1970.1054406>.
- Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. Boolq: Exploring the surprising difficulty of natural yes/no questions. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational*

- Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, page 29242936. Association for Computational Linguistics, 2019. doi: 10.18653/v1/n19-1300. URL <http://dx.doi.org/10.18653/v1/n19-1300>.
- Tri Dao, Dan Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. Flashattention: Fast and memory-efficient exact attention with io-awareness. In *Advances in Neural Information Processing Systems 35*, NeurIPS 2022, page 1634416359. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2022. doi: 10.52202/068431-1189. URL <http://dx.doi.org/10.52202/068431-1189>.
- Michiel de Jong, Yury Zemlyanskiy, Nicholas FitzGerald, Joshua Ainslie, Sumit Sanghai, Fei Sha, and William Cohen. Pre-computed memory or on-the-fly encoding? a hybrid approach to retrieval augmentation makes the most of your compute, 2023. URL <https://arxiv.org/abs/2301.10448>.
- Yuntian Deng, Kiran Prasad, Roland Fernandez, Paul Smolensky, Vishrav Chaudhary, and Stuart Shieber. Implicit chain of thought reasoning via knowledge distillation, 2023. URL <https://arxiv.org/abs/2311.01460>.
- Yuntian Deng, Yejin Choi, and Stuart Shieber. From explicit cot to implicit cot: Learning to internalize cot step by step, 2024. URL <https://arxiv.org/abs/2405.14838>.
- Shrey Desai and Greg Durrett. Calibration of pre-trained transformers. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, page 295302. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.emnlp-main.21. URL <http://dx.doi.org/10.18653/v1/2020.emnlp-main.21>.
- Raúl Díaz and Amit Marathe. Soft labels for ordinal regression. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, page 47334742. IEEE, 2019. doi: 10.1109/cvpr.2019.00487. URL <http://dx.doi.org/10.1109/cvpr.2019.00487>.
- Edward S. Epstein. A scoring system for probability forecasts of ranked categories. *Journal of Applied Meteorology*, 8(6):985987, December 1969. ISSN 0021-8952. doi: 10.1175/1520-0450(1969)008<0985:assfpf>2.0.co;2. URL [http://dx.doi.org/10.1175/1520-0450\(1969\)008<0985:assfpf>2.0.co;2](http://dx.doi.org/10.1175/1520-0450(1969)008<0985:assfpf>2.0.co;2).
- Guhao Feng, Bohang Zhang, Yuntian Gu, Haotian Ye, Di He, and Liwei Wang. Towards revealing the mystery behind chain of thought: A theoretical perspective, 2023. URL <https://arxiv.org/abs/2305.15408>.
- Eibe Frank and Mark Hall. *A Simple Approach to Ordinal Classification*, page 145156. Springer Berlin Heidelberg, 2001. ISBN 9783540447955. doi: 10.1007/3-540-44795-4_13. URL http://dx.doi.org/10.1007/3-540-44795-4_13.
- Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks, 2017. URL <https://arxiv.org/abs/1705.08500>.
- Yonatan Geifman and Ran El-Yaniv. Selectivenet: A deep neural network with an integrated reject option, 2019. URL <https://arxiv.org/abs/1901.09192>.
- Saibo Geng, Martin Josifoski, Maxime Peyrard, and Robert West. Grammar-constrained decoding for structured nlp tasks without finetuning. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, page 1093210952. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.emnlp-main.674. URL <http://dx.doi.org/10.18653/v1/2023.emnlp-main.674>.
- In Gim, Guojun Chen, Seung-seob Lee, Nikhil Sarda, Anurag Khandelwal, and Lin Zhong. Prompt cache: Modular attention reuse for low-latency inference, 2023. URL <https://arxiv.org/abs/2311.04934>.
- Tilmann Gneiting and Adrian E Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359378, March 2007. ISSN 1537-274X. doi: 10.1198/016214506000001437. URL <http://dx.doi.org/10.1198/016214506000001437>.

- Sachin Goyal, Ziwei Ji, Ankit Singh Rawat, Aditya Krishna Menon, Sanjiv Kumar, and Vaishnavh Nagarajan. Think before you speak: Training language models with pause tokens, 2023. URL <https://arxiv.org/abs/2310.02226>.
- Andrey Gromov, Kushal Tirumala, Hassan Shapourian, Paolo Gloriosi, and Daniel A. Roberts. The unreasonable ineffectiveness of the deeper layers, 2024. URL <https://arxiv.org/abs/2403.17887>.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks, 2017. URL <https://arxiv.org/abs/1706.04599>.
- Kishaloy Halder, Alan Akbik, Josip Krapac, and Roland Vollgraf. Task-aware representation of sentences for generic text classification. In *Proceedings of the 28th International Conference on Computational Linguistics*, page 32023213. International Committee on Computational Linguistics, 2020. doi: 10.18653/v1/2020.coling-main.285. URL <http://dx.doi.org/10.18653/v1/2020.coling-main.285>.
- Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space, 2024. URL <https://arxiv.org/abs/2412.06769>.
- Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. 2016. doi: 10.48550/ARXIV.1610.02136. URL <https://arxiv.org/abs/1610.02136>.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding, 2020. URL <https://arxiv.org/abs/2009.03300>.
- John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, page 27332743. Association for Computational Linguistics, 2019. doi: 10.18653/v1/d19-1275. URL <http://dx.doi.org/10.18653/v1/d19-1275>.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network, 2015. URL <https://arxiv.org/abs/1503.02531>.
- Cheng-Ping Hsieh, Simeng Sun, Samuel Krman, Shantanu Acharya, Dima Rekeshe, Fei Jia, Yang Zhang, and Boris Ginsburg. Ruler: What’s the real context size of your long-context language models?, 2024. URL <https://arxiv.org/abs/2404.06654>.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021. URL <https://arxiv.org/abs/2106.09685>.
- Samuel Humeau, Kurt Shuster, Marie-Anne Lachaux, and Jason Weston. Poly-encoders: Transformer architectures and pre-training strategies for fast and accurate multi-sentence scoring, 2019. URL <https://arxiv.org/abs/1905.01969>.
- Dieuwke Hupkes, Verna Dankers, Mathijs Mul, and Elia Bruni. Compositionality decomposed: How do neural networks generalise? *Journal of Artificial Intelligence Research*, 67:757795, April 2020. ISSN 1076-9757. doi: 10.1613/jair.1.11674. URL <http://dx.doi.org/10.1613/jair.1.11674>.
- Jordan Juravsky, Bradley Brown, Ryan Ehrlich, Daniel Y. Fu, Christopher Ré, and Azalia Mirhoseini. Hydragen: High-throughput llm inference with shared prefixes, 2024. URL <https://arxiv.org/abs/2402.05099>.
- Daniel Keysers, Nathanael Schärli, Nathan Scales, Hylke Buisman, Daniel Furrer, Sergii Kashubin, Nikola Momchev, Danila Sinopalnikov, Lukasz Stafiniak, Tibor Tihon, Dmitry Tsarkov, Xiao Wang, Marc van Zee, and Olivier Bousquet. Measuring compositional generalization: A comprehensive method on realistic data, 2019. URL <https://arxiv.org/abs/1912.09713>.

- Omar Khattab and Matei Zaharia. Colbert: Efficient and effective passage search via contextualized late interaction over bert, 2020. URL <https://arxiv.org/abs/2004.12832>.
- Najoung Kim and Tal Linzen. Cogs: A compositional generalization challenge based on semantic interpretation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, page 90879105. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.emnlp-main.731. URL <http://dx.doi.org/10.18653/v1/2020.emnlp-main.731>.
- Yoon Kim and Alexander M. Rush. Sequence-level knowledge distillation. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, page 13171327. Association for Computational Linguistics, 2016. doi: 10.18653/v1/d16-1139. URL <http://dx.doi.org/10.18653/v1/d16-1139>.
- Ananya Kumar, Percy Liang, and Tengyu Ma. Verified uncertainty calibration, 2019. URL <https://arxiv.org/abs/1909.10155>.
- Ananya Kumar, Aditi Raghunathan, Robbie Jones, Tengyu Ma, and Percy Liang. Fine-tuning can distort pretrained features and underperform out-of-distribution, 2022. URL <https://arxiv.org/abs/2202.10054>.
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with pagedattention. In *Proceedings of the 29th Symposium on Operating Systems Principles, SOSP 23*, page 611626. ACM, October 2023. doi: 10.1145/3600006.3613165. URL <http://dx.doi.org/10.1145/3600006.3613165>.
- Brenden M. Lake and Marco Baroni. Generalization without systematicity: On the compositional skills of sequence-to-sequence recurrent networks. 2017. doi: 10.48550/ARXIV.1711.00350. URL <https://arxiv.org/abs/1711.00350>.
- Stefan Larson, Anish Mahendran, Joseph J. Peper, Christopher Clarke, Andrew Lee, Parker Hill, Jonathan K. Kummerfeld, Kevin Leach, Michael A. Laurenzano, Lingjia Tang, and Jason Mars. An evaluation dataset for intent classification and out-of-scope prediction. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, page 13111316. Association for Computational Linguistics, 2019. doi: 10.18653/v1/d19-1131. URL <http://dx.doi.org/10.18653/v1/d19-1131>.
- Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding, 2022. URL <https://arxiv.org/abs/2211.17192>.
- Mosh Levy, Alon Jacoby, and Yoav Goldberg. Same task, more tokens: the impact of input length on the reasoning performance of large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 1533915353. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.818. URL <http://dx.doi.org/10.18653/v1/2024.acl-long.818>.
- Zhiyuan Li, Hong Liu, Denny Zhou, and Tengyu Ma. Chain of thought empowers transformers to solve inherently serial problems, 2024. URL <https://arxiv.org/abs/2402.12875>.
- Zongxi Li, Xianming Li, Yuzhang Liu, Haoran Xie, Jing Li, Fu-lee Wang, Qing Li, and Xiaoqin Zhong. Label supervised llama finetuning, 2023. URL <https://arxiv.org/abs/2310.01208>.
- Tom Lieberum, Matthew Rahtz, János Kramár, Neel Nanda, Geoffrey Irving, Rohin Shah, and Vladimir Mikulik. Does circuit analysis interpretability scale? evidence from multiple choice capabilities in chinchilla, 2023. URL <https://arxiv.org/abs/2307.09458>.
- Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 32143252. Association for Computational Linguistics, 2022. doi: 10.18653/v1/2022.acl-long.229. URL <http://dx.doi.org/10.18653/v1/2022.acl-long.229>.

- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157173, 2024. ISSN 2307-387X. doi: 10.1162/tac1_a_00638. URL http://dx.doi.org/10.1162/tac1_a_00638.
- Weitang Liu, Xiaoyun Wang, John D. Owens, and Yixuan Li. Energy-based out-of-distribution detection, 2020. URL <https://arxiv.org/abs/2010.03759>.
- Yuxiang Lu, Yiding Liu, Jiaxiang Liu, Yunsheng Shi, Zhengjie Huang, Shikun Feng Yu Sun, Hao Tian, Hua Wu, Shuaiqiang Wang, Dawei Yin, and Haifeng Wang. Ernie-search: Bridging cross-encoder with dual-encoder via self on-the-fly distillation for dense passage retrieval, 2022. URL <https://arxiv.org/abs/2205.09153>.
- Yun Luo, Zhen Yang, Fandong Meng, Yafu Li, Jie Zhou, and Yue Zhang. An empirical study of catastrophic forgetting in large language models during continual fine-tuning. *IEEE Transactions on Audio, Speech and Language Processing*, 33:37763786, 2025. ISSN 2998-4173. doi: 10.1109/taslpro.2025.3606231. URL <http://dx.doi.org/10.1109/taslpro.2025.3606231>.
- Sean MacAvaney, Franco Maria Nardini, Raffaele Perego, Nicola Tonellotto, Nazli Goharian, and Ophir Frieder. Efficient document re-ranking for transformers by precomputing term representations. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR 20, page 4958. ACM, 2020. doi: 10.1145/3397271.3401093. URL <http://dx.doi.org/10.1145/3397271.3401093>.
- Peter McCullagh. Regression models for ordinal data. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 42(2):109127, January 1980. ISSN 1467-9868. doi: 10.1111/j.2517-6161.1980.tb01109.x. URL <http://dx.doi.org/10.1111/j.2517-6161.1980.tb01109.x>.
- Xin Men, Mingyu Xu, Qingyu Zhang, Qianhao Yuan, Bingning Wang, Hongyu Lin, Yaojie Lu, Xianpei Han, and Weipeng Chen. Shortgpt: Layers in large language models are more redundant than you expect. In *Findings of the Association for Computational Linguistics: ACL 2025*, page 2019220204. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.findings-acl.1035. URL <http://dx.doi.org/10.18653/v1/2025.findings-acl.1035>.
- William Merrill and Ashish Sabharwal. The parallelism tradeoff: Limitations of log-precision transformers. *Transactions of the Association for Computational Linguistics*, 11:531545, 2023a. ISSN 2307-387X. doi: 10.1162/tac1_a_00562. URL http://dx.doi.org/10.1162/tac1_a_00562.
- William Merrill and Ashish Sabharwal. The expressive power of transformers with chain of thought, 2023b. URL <https://arxiv.org/abs/2310.07923>.
- Marius Mosbach, Tiago Pimentel, Shauli Ravfogel, Dietrich Klakow, and Yanai Elazar. Few-shot fine-tuning vs. in-context learning: A fair comparison and evaluation. In *Findings of the Association for Computational Linguistics: ACL 2023*, page 1228412314. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.findings-acl.779. URL <http://dx.doi.org/10.18653/v1/2023.findings-acl.779>.
- ALLAN H. MURPHY. The ranked probability score and the probability score: A comparison^{>1,}</sup>². *Monthly Weather Review*, 98(12):917924, December 1970. ISSN 1520-0493. doi: 10.1175/1520-0493(1970)098<0917:trpsat>2.3.co;2. URL [http://dx.doi.org/10.1175/1520-0493\(1970\)098<0917:trpsat>2.3.co;2](http://dx.doi.org/10.1175/1520-0493(1970)098<0917:trpsat>2.3.co;2).
- Rafael Müller, Simon Kornblith, and Geoffrey Hinton. When does label smoothing help?, 2019. URL <https://arxiv.org/abs/1906.02629>.
- Yixin Nie, Adina Williams, Emily Dinan, Mohit Bansal, Jason Weston, and Douwe Kiela. Adversarial nli: A new benchmark for natural language understanding. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, page 48854901. Association for Computational Linguistics, 2020a. doi: 10.18653/v1/2020.acl-main.441. URL <http://dx.doi.org/10.18653/v1/2020.acl-main.441>.

- Yixin Nie, Xiang Zhou, and Mohit Bansal. What can we learn from collective human opinions on natural language inference data? In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, page 91319143. Association for Computational Linguistics, 2020b. doi: 10.18653/v1/2020.emnlp-main.734. URL <http://dx.doi.org/10.18653/v1/2020.emnlp-main.734>.
- Zhenxing Niu, Mo Zhou, Le Wang, Xinbo Gao, and Gang Hua. Ordinal regression with multiple output cnn for age estimation. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, page 49204928. IEEE, 2016. doi: 10.1109/cvpr.2016.532. URL <http://dx.doi.org/10.1109/cvpr.2016.532>.
- Rodrigo Nogueira and Kyunghyun Cho. Passage re-ranking with bert, 2019. URL <https://arxiv.org/abs/1901.04085>.
- Mahdi Pakdaman Naeini, Gregory Cooper, and Milos Hauskrecht. Obtaining well calibrated probabilities using bayesian binning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 29(1), February 2015. ISSN 2159-5399. doi: 10.1609/aaai.v29i1.9602. URL <http://dx.doi.org/10.1609/aaai.v29i1.9602>.
- Koyena Pal, Jiuding Sun, Andrew Yuan, Byron Wallace, and David Bau. Future lens: Anticipating subsequent tokens from a single hidden state. In *Proceedings of the 27th Conference on Computational Natural Language Learning (CoNLL)*, page 548560. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.conll-1.37. URL <http://dx.doi.org/10.18653/v1/2023.conll-1.37>.
- Nikolaos Pappas and James Henderson. Gile: A generalized input-label embedding for text classification. *Transactions of the Association for Computational Linguistics*, 7:139155, November 2019. ISSN 2307-387X. doi: 10.1162/tac1_a_00259. URL http://dx.doi.org/10.1162/tac1_a_00259.
- Pouya Pezeshkpour and Estevam Hruschka. Large language models sensitivity to the order of options in multiple-choice questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*, page 20062017. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.findings-naacl.130. URL <http://dx.doi.org/10.18653/v1/2024.findings-naacl.130>.
- Jacob Pfau, William Merrill, and Samuel R. Bowman. Let’s think dot by dot: Hidden computation in transformer language models, 2024. URL <https://arxiv.org/abs/2404.15758>.
- Reiner Pope, Sholto Douglas, Aakanksha Chowdhery, Jacob Devlin, James Bradbury, Anselm Levskaya, Jonathan Heek, Kefan Xiao, Shivani Agrawal, and Jeff Dean. Efficiently scaling transformer inference, 2022. URL <https://arxiv.org/abs/2211.05102>.
- Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, page 39803990. Association for Computational Linguistics, 2019. doi: 10.18653/v1/d19-1410. URL <http://dx.doi.org/10.18653/v1/d19-1410>.
- Nithya Sambasivan, Shivani Kapania, Hannah Highfill, Diana Akrong, Praveen Paritosh, and Lora M Aroyo. everyone wants to do the model work, not the data work: Data cascades in high-stakes ai. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, CHI 21, page 115. ACM, May 2021. doi: 10.1145/3411764.3445518. URL <http://dx.doi.org/10.1145/3411764.3445518>.
- Roy Schwartz, Gabriel Stanovsky, Swabha Swayamdipta, Jesse Dodge, and Noah A. Smith. The right tool for the job: Matching model and instance complexities. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, page 66406651. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.acl-main.593. URL <http://dx.doi.org/10.18653/v1/2020.acl-main.593>.
- Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed Chi, Nathanael Schärli, and Denny Zhou. Large language models can be easily distracted by irrelevant context, 2023a. URL <https://arxiv.org/abs/2302.00093>.

- Xintong Shi, Wenzhi Cao, and Sebastian Raschka. Deep neural networks for rank-consistent ordinal regression based on conditional probabilities. *Pattern Analysis and Applications*, 26(3):941955, 2023b. ISSN 1433-755X. doi: 10.1007/s10044-023-01181-9. URL <http://dx.doi.org/10.1007/s10044-023-01181-9>.
- Charlie Snell, Dan Klein, and Ruiqi Zhong. Learning by distilling context, 2022. URL <https://arxiv.org/abs/2209.15189>.
- Zayne Sprague, Fangcong Yin, Juan Diego Rodriguez, Dongwei Jiang, Manya Wadhwa, Prasann Singhal, Xinyu Zhao, Xi Ye, Kyle Mahowald, and Greg Durrett. To cot or not to cot? chain-of-thought helps mainly on math and symbolic reasoning, 2024. URL <https://arxiv.org/abs/2409.12183>.
- Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, page 28182826. IEEE, 2016. doi: 10.1109/cvpr.2016.308. URL <http://dx.doi.org/10.1109/cvpr.2016.308>.
- Zhi Rui Tam, Cheng-Kuang Wu, Yi-Lin Tsai, Chieh-Yen Lin, Hung-yi Lee, and Yun-Nung Chen. Let me speak freely? a study on the impact of format restrictions on large language model performance. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, page 12181236. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.emnlp-industry.91. URL <http://dx.doi.org/10.18653/v1/2024.emnlp-industry.91>.
- Ian Tenney, Dipanjan Das, and Ellie Pavlick. Bert rediscovers the classical nlp pipeline. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, page 45934601. Association for Computational Linguistics, 2019. doi: 10.18653/v1/p19-1452. URL <http://dx.doi.org/10.18653/v1/p19-1452>.
- Xinpeng Wang, Bolei Ma, Chengzhi Hu, Leon Weber-Genzel, Paul Röttger, Frauke Kreuter, Dirk Hovy, and Barbara Plank. my answer is c: First-token probabilities do not match text answers in instruction-tuned language models. In *Findings of the Association for Computational Linguistics ACL 2024*, page 74077416. Association for Computational Linguistics, 2024a. doi: 10.18653/v1/2024.findings-acl.441. URL <http://dx.doi.org/10.18653/v1/2024.findings-acl.441>.
- Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhui Chen. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. In *Advances in Neural Information Processing Systems 37*, NeurIPS 2024, page 9526695290. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2024b. doi: 10.52202/079017-3018. URL <http://dx.doi.org/10.52202/079017-3018>.
- Ziming Wang, Zeyu Shi, Haoyi Zhou, Shiqi Gao, Qingyun Sun, and Jianxin Li. Towards objective fine-tuning: How llms prior knowledge causes potential poor calibration? In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 1483014853. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.acl-long.722. URL <http://dx.doi.org/10.18653/v1/2025.acl-long.722>.
- Benjamin Warner, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom Aarsen, Griffin Thomas Adams, Jeremy Howard, and Iacopo Poli. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 25262547. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.acl-long.127. URL <http://dx.doi.org/10.18653/v1/2025.acl-long.127>.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems 35*, NeurIPS 2022, page 2482424837. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2022. doi: 10.52202/068431-1800. URL <http://dx.doi.org/10.52202/068431-1800>.

- Brandon T. Willard and Rémi Louf. Efficient guided generation for large language models, 2023. URL <https://arxiv.org/abs/2307.09702>.
- Adina Williams, Nikita Nangia, and Samuel Bowman. A broad-coverage challenge corpus for sentence understanding through inference. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, page 11121122. Association for Computational Linguistics, 2018. doi: 10.18653/v1/n18-1101. URL <http://dx.doi.org/10.18653/v1/n18-1101>.
- Hugh Mee Wong, Rick Nouwen, and Albert Gatt. When models decide and when they bind: A two-stage computation for multiple-choice question answering. In *Findings of the Association for Computational Linguistics: ACL 2026*, page 2280422821. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.findings-acl.1144. URL <http://dx.doi.org/10.18653/v1/2026.findings-acl.1144>.
- Wilson Wu, John X. Morris, and Lionel Levine. Do language models plan ahead for future tokens?, 2024. URL <https://arxiv.org/abs/2404.00859>.
- Mengge Xue, Zhenyu Hu, Liqun Liu, Kuo Liao, Shuang Li, Honglin Han, Meng Zhao, and Chengguo Yin. Strengthened symbol binding makes large language models reliable multiple-choice selectors. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 43314344. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.237. URL <http://dx.doi.org/10.18653/v1/2024.acl-long.237>.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report, 2025a. URL <https://arxiv.org/abs/2505.09388>.
- Zhen Yang, Ping Jian, and Chengzhi Li. Option symbol matters: Investigating and mitigating multiple-choice option symbol bias of large language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, page 19021917. Association for Computational Linguistics, 2025b. doi: 10.18653/v1/2025.naacl-long.95. URL <http://dx.doi.org/10.18653/v1/2025.naacl-long.95>.
- Wenpeng Yin, Jamaal Hay, and Dan Roth. Benchmarking zero-shot text classification: Datasets, evaluation and entailment approach. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, page 39123921. Association for Computational Linguistics, 2019. doi: 10.18653/v1/d19-1404. URL <http://dx.doi.org/10.18653/v1/d19-1404>.
- Li Yuan, Francis EH Tay, Guilin Li, Tao Wang, and Jiashi Feng. Revisiting knowledge distillation via label smoothing regularization. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, page 39023910. IEEE, 2020. doi: 10.1109/cvpr42600.2020.00396. URL <http://dx.doi.org/10.1109/cvpr42600.2020.00396>.
- Urchade Zaratiana, Nadi Tomeh, Pierre Holat, and Thierry Charnois. Gliner: Generalist model for named entity recognition using bidirectional transformer. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, page 53645376. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.naacl-long.300. URL <http://dx.doi.org/10.18653/v1/2024.naacl-long.300>.
- Urchade Zaratiana, Gil Pasternak, Oliver Boyd, George Hurn-Maloney, and Ash Lewis. Gliner2: An efficient multi-task information extraction system with schema-driven interface, 2025. URL <https://arxiv.org/abs/2507.18546>.

- Zhilu Zhang and Mert R. Sabuncu. Self-distillation as instance-specific label smoothing. 2020. doi: 10.48550/ARXIV.2006.05065. URL <https://arxiv.org/abs/2006.05065>.
- Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. Large language models are not robust multiple choice selectors, 2023. URL <https://arxiv.org/abs/2309.03882>.
- Lianmin Zheng, Liangsheng Yin, Zhiqiang Xie, Chuyue Sun, Jeff Huang, Cody Yu, Shiyi Cao, Christos Kozyrakis, Ion Stoica, Joseph Gonzalez, Clark Barrett, and Ying Sheng. Sglang: Efficient execution of structured language model programs. In *Advances in Neural Information Processing Systems 37*, NeurIPS 2024, page 6255762583. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2024. doi: 10.52202/079017-2000. URL <http://dx.doi.org/10.52202/079017-2000>.

A Appendix

A.1 Additional ablation tables and figures

Tables 8 and 9 give the full numbers for the candidate-blind closure (Section 6.2) and the rendered-null-line ablation (Section 6.4). Figure 5 gives the long-state truncation histogram behind Section 4.5. The serving-path engineering behind Section 7.5, with Figure 7, is in Appendix A.3.

Table 8: Candidate-blind Z , closure at full depth vs. the original tap-20 student.

	Tap 20	Full depth (28)	Teacher
MMLU-Pro among- K / choices-only / shuffled-q	.157 / .166 / .182	.147 / .177 / .156	.310 / .222 / .208
Δ_q (choices-only) / Δ_q^{sh} (primary)	-.009 / -.026	-.030 / -.008	.088 / .102
SNLI / MNLI / ANLI / BoolQ	.905/.873/.541/.826	.881/.807/.462/.712	-

Table 9: Rendered $\emptyset$ line vs. the factored null, matched checkpoints (tap 20).

	Rendered $\emptyset$ line	Factored null (no rendered line)
MMLU-Pro false-abstain rate	.692	.003
MMLU-Pro accuracy	.133	.353
Banking77 (unseen labels) accuracy / false-abstain	.063 / .924	.579 / .135
CLINC-OOS accuracy / null AUROC	.639 / .840	.798 / .973
Δ_q^{sh} (unchanged)	.118	.122

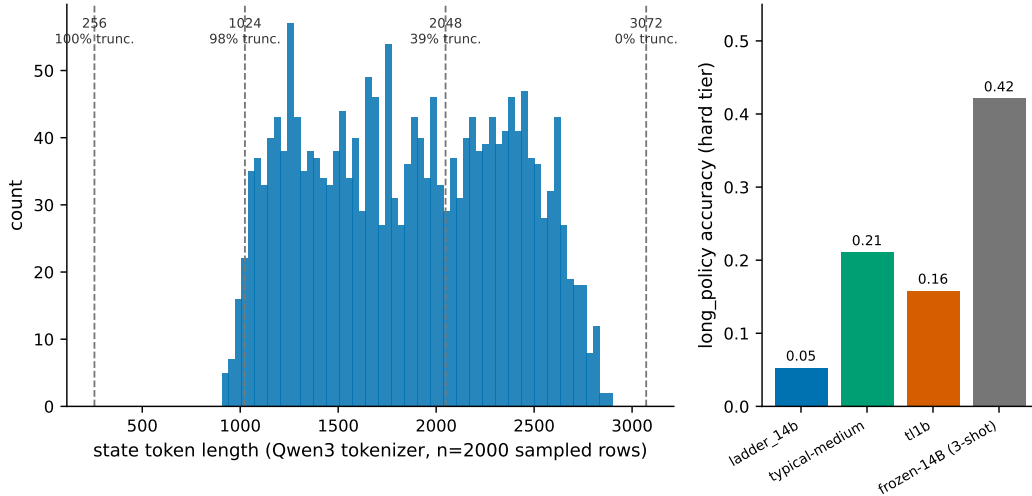


Figure 5: Left: histogram of `data_wf_long` state token lengths with truncation fractions at 256/1024/2048/3072 tokens. Right: `long_policy` (hard-tier) accuracy across the long-state bug/fix sequence.

A.2 Native choice vs. energy scoring vs. batched log-probability, full K -sweep

At $M=256$ queries against one cached 256-token state, native choice’s per-query marginal cost is 1.07, 1.55, 3.38, and 43.8 ms at $K=2, 10, 32, 256$; a candidate-blind energy-scoring path is flat at ≈ 1 ms across the same range; and a baseline that re-runs the backbone once per candidate costs 2.1, 4.6, 12.2, and 101.7 ms. These values are transcribed from the run’s log, because the run’s `bench.json` was lost to an out-of-memory crash. Batched marginal cost at $M=256$ scales more steeply with both size and K than single-decision cost, from 2.7 ms at 1.7B/ $K=2$ to 110 ms at 14B/ $K=256$, so an energy-scoring front end gains value as the backbone grows.

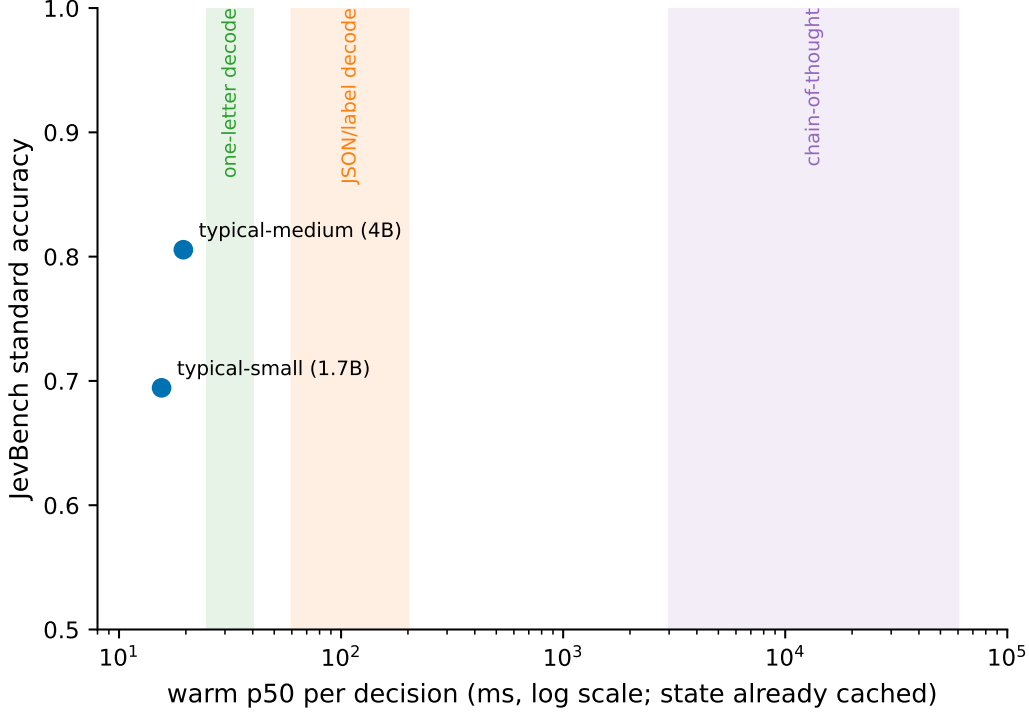


Figure 6: **Public subset, unranked.** JevBench standard accuracy vs. *warm* per-decision p50 latency (ms, log scale) for the v1 release checkpoints (ts1b, tm1b), with generative-LLM decode-latency comparison bands. Latency is the served path with the state prefix already cached (runs/serve_bench2, $K=2$, 256-token state, one stream, in process, model load excluded), the same quantity as Appendix A.3. The single-decision ladder of Table 26 measures a different quantity: it includes the state encode on the unoptimized path. Referenced from Section 7.5.

A.3 Serving-path engineering

The public inference package initially deep-copied the entire cached prefix’s key-value tensors on every decision. Replacing this with a zero-copy strided view plus cached attention masks and position tensors reduced warm per-decision latency from 21–27 ms to **15.5–17 ms** (typical-small v1, Qwen3-1.7B) and **19–21 ms** (typical-medium v1, Qwen3-4B) on one H100 (Figure 7). tm2 (Qwen3.5-4B, typical-medium v2) serves at 34–46 ms warm p50 on the same package with reference DeltaNet kernels. `torch.compile/CUDA`-graph capture reached 8 ms at a single fixed input shape, but probabilities shifted by up to .1 across the shape buckets the serving path sees, and the mutable key-value cache produced stale-buffer crashes under the standard capture-and-replay pattern, so we reverted it. Manual per-bucket graph capture is the remaining path toward ~ 10 ms decisions.

A.4 Mixture sweep, full table

Table 10: Mixture sweep at 1.7B, matched steps, referenced from Section 7.2. “6A” is the $E:0.35/K:0.25/W:0.40$ point; “e45+null” adds `null_aug W:0.20` to $E:0.45/K:0.20/W:0.35$.

	Base (E only)	6A	e45	e45+null	long _{e45} (1024 tok)
CLINC-150 acc / false-abstain	.845/.05	.734/.17	.779/.10	.805/.08	.741/.13
MMLU-Pro among- K / Δ_q^{sh}	.353/.122	.330/.119	.338/.115	.353/.143	.329/.116
Held-out score NLL	–	2.03	1.52	2.04	1.26
JevBench std / hard	.694/.378	.750/.387	.833/.396	.806/.396	.806/ .450

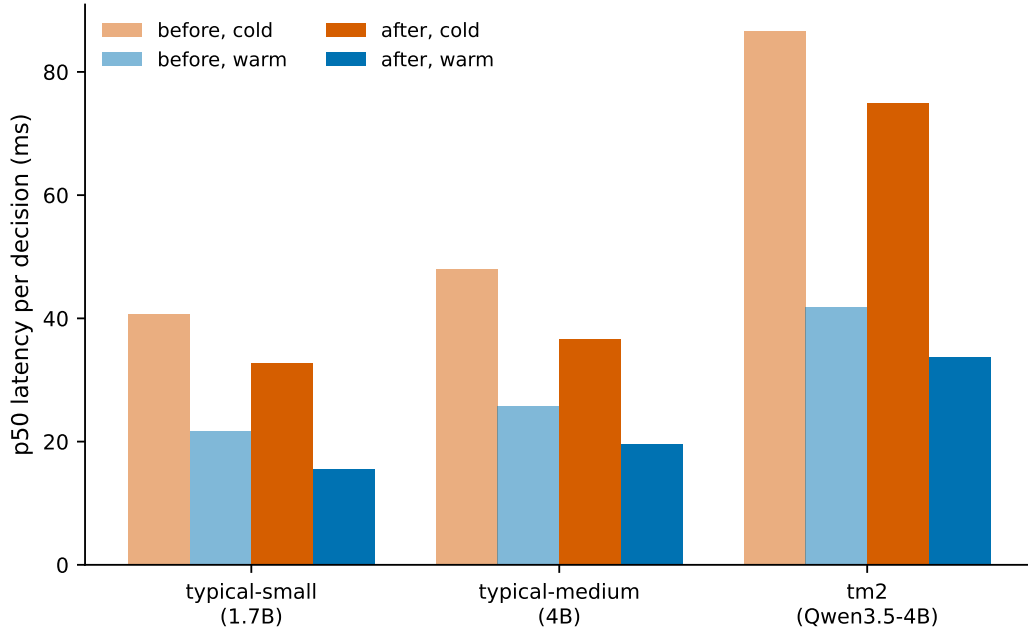


Figure 7: Per-decision p50 latency before/after the zero-copy prefix fix, cold vs. warm, for typical-small v1 (1.7B), typical-medium v1 (Qwen3-4B), and tm2 (Qwen3.5-4B, typical-medium v2), at $K=2$, 256-token state.

Before the facts-first fix (Section 4.5), the long-state row (long_{e45} , 1,024 tokens, 25% long-policy rows) alone raised `long_policy` from .16 to .37 and `multi_hop` from .17 to .28 relative to 6A, and gave the best soft-target NLLs of the sweep.

A.5 Typed primitives: parameterization, routing, and full results

Choice is the K -way readout of Section 3.2, unchanged: a hard one-of- K target trained with soft cross-entropy against Equation (2). **Score** is ordinal (severity, quality, urgency). Plain Choice penalizes a one-level miss as much as an opposite-end miss, so Score uses an ordinal-smoothed target,

$$\tilde{t}_j \propto \exp\left(-\frac{|j-y|}{\tau}\right), \quad \tau = 0.7, \quad (3)$$

normalized over valid levels, with $\tau=0$ recovering one-hot Choice. This is a pure target change with no architecture change. **Noul** is a yes/no proposition. Rendered as 2-way Choice it has no order-invariance guarantee, since “yes” and “no” occupy two different suffix positions with two different contextual states. Noul therefore gets a dedicated Bernoulli head that reads the query alone, with no candidate text,

$$P(\text{yes}) = \sigma(w_n \cdot h_D), \quad (4)$$

which is exactly order-invariant by construction and verified as an exact-zero reversed-label control. The remaining mass $1-P(\text{yes})$ spreads over the other candidate(s), and the factored-null gate still applies.

Per-row routing. A row is Noul-eligible iff its candidate set is exactly $\{\text{yes}, \text{no}\}$; every other row scores as ordinary Choice, bit-identical to a run with Noul disabled. An early release candidate that applied the Bernoulli head at the *model* level collapsed to chance, because Choice- and Score-typed rows in the same batch were then scored with no candidate text at all. Routing per row (verified by 140 unit tests) restored it. The type is a contract on the input, and breaking that contract collapses accuracy to chance.

Table 11: Typed primitives vs. their K -way Choice control, matched fine-tunes, referenced from Section 7.1.

Score	K -way Choice	Ordinal-smoothed ($\tau=0.7$)
Held-out urgency: acc / NLL / ECE	.505 / 2.07 / .35	.508 / 1.23 / .19
JevBench hard: acc / Brier	.360 / .88	.459 / .70
JevBench ordinal items (12), decisions changed	–	0 of 12
Noul	2-way Choice	Bernoulli
PagerDuty (floor .792): acc / NLL	.602 / 1.04	.886 / .28
Reversed-label max $ \Delta P(\text{yes}) $	.55	0 (exact, all sets)

A.6 DecisionMix v2, full numbers

Held-out family/grammar/style accuracy moves from .48/.51/.49 (matched control, never trained on data_wh) to .83/.88/.89 (DecisionMix v2), and rubric-flip accuracy moves from .48 to .71, measured within the generator’s own rule engine (Section 4.2). The uncertainty corpus moves typed-decisions NLL from 2.06 to 1.71 (Section 4.3). Level-7 composition is in Appendix A.9.

A.7 Full hyperparameters

Table 12 gives the exact configuration of each checkpoint in the paper. All runs use AdamW, bf16, and LoRA rank 16 on the top eight kept layers of the truncated backbone; the family-balanced sampler and null augmentation are described in Sections 4.1 and 4.4.

Table 12: Training configuration by checkpoint. Tap is given as layer / total layers of the base model. $E:K:W:U$ is the family-sampler weight per batch. Max state is the training-time token budget for the cached prefix.

Checkpoint	Backbone	Tap	Steps	Batch	$E:K:W:U$	Max state
typical-small-preview	Qwen3-1.7B-Base	20/28	12,000	64	.35:.25:.40:0	256
ts1b (typical-small v1)	Qwen3-1.7B-Base	20/28	12,000	64	.40:.15:.35:.10	1,024
tm1b (typical-medium v1)	Qwen3-4B-Base	26/36	12,000	64	.40:.15:.35:.10	1,024
ladder_8b	Qwen3-8B-Base	26/36	12,000	64	.35:.25:.40:0	256
ladder_14b	Qwen3-14B-Base	28/40	12,000	64	.35:.25:.40:0	256
t11b (14B)	Qwen3-14B-Base	28/40	8,000	64	DecisionMix v2 + U , facts-first	3,072
tm2 (typical-medium v2)	Qwen3.5-4B-Base	$\approx 71\%$	12,000	64	.40:.15:.35:.10	2,048

ts1b/tm1b additionally set null_aug W:0.20, ordinal_smooth 0.7 (Score), noul_head bern (routed per row, Section 3.5), null factored, and nc_render letters_nonull. t11b adds -drop_truncated, frozen-teacher KD ($\alpha \cdot \text{CE} + \beta \cdot \text{KL}$, $T=2$, against the same backbone’s own frozen 3-shot letter distribution on 63k labelled rows), and checkpoint selection on held-out uncertainty-plus-curriculum validation NLL (-best_on) in place of in-distribution loss. Final fitted temperature and null offset for ts1b: $T=1.124$, offset 0 (post-hoc, applied at eval). ts1c (typical-small v2) is the 1.7B model trained on the facts-first long-state corpus; its long-state and JevBench numbers are in Table 21.

A.8 Corpus construction detail

data_v5 (633,213 train / 4,001 val rows) covers evidence tasks and classification with a deliberately wide span of label vocabularies at low rows-per-source, so the model cannot memorize a fixed small set of label sets and must read the rendered options. data_kb (147,447 train / 3,001 val) is distilled from ARC-Easy/Challenge, OpenBookQA, CommonsenseQA, SciQ, QASC, LogiQA, AQUA-RAT, MedMCQA, and MMLU-auxiliary-train, with TruthfulQA-MC1 held out entirely and two items excluded from all MMLU-Pro numbers after a leakage audit against that evaluation set. data_wf (72,006 train / 3,000 val) is our own rubric-conditioned workflow generator, 15 families of Noul/Choice/Score decisions (policy checks, adequacy, support, fact verification, eligibility, routing, action selection, enumeration extraction, categorical judgment, tool selection, and four ordinal quality/severity/relevance/completeness scales, plus urgency); three whole families (el-

igibility, tool_select, urgency, one per type) are held out of training entirely, and 26,513 of the 72,006 rows belong to *rubric groups*: the same state and candidate set rendered under two or three different rubrics with different gold answers, built to measure and train rubric dependence over state/candidate pattern-matching. **data_wf_hf** (86,600 train rows) adds three externally sourced sets in the same typed shape: *cua-s1-forms* (60,000 rows, procedural GUI-form option-scoring, MIT license, with a form-signature-disjoint 24,370-row test split and a 196-row real-usage demo split held out), *systemone-lite-general* (20,000 rows, rule-labeled typed decisions, MIT, with a 5,400-row hard split held out), and *jev-4b-distill-data* (6,600 rows, programmatic gold, Apache-2.0, with a 600-row adversarial split held out). **data_wf_long** (23,318 train rows) is a longer-state variant of the workflow generator for multi-paragraph policy documents, rendered facts-first (Section 4.5). **data_wh** (60,605 train / 3,000 val) is a second, independent rubric-conditioned generator (DecisionMix v2): a programmatic rule engine over 12 domains and six boolean conditions at seven levels of compositional depth, where levels 1–6 are trained and level 7 (compositions requiring temporal, unit-conversion, expected-value, or trade-off reasoning that the generator does not otherwise produce) is evaluation-only; every trained row belongs to a counterfactual rubric group. One whole domain/level pair, three whole domains, and two rubric styles are held out. **data_u** (31,029 train / 2,000 val) is a soft-target uncertainty corpus: the validation split of UNLI, the ambiguous rows of AmbiEnt (uniform target over the listed labels), and four synthetic generators with exact closed-form posteriors (partial-evidence combination, a noisy-sensor Bayesian update, ordinal confusion via an inverted confusion matrix, and conflicting-sources log-odds pooling), each with a held-out parameter range; ChaosNLI-ambiguity (MNLI portion) is evaluation-only.

Table 13: **data_wf**: 15 families, 72,006 train rows, 3 held out (marked *). “In rubric groups” counts rows that are part of a same-state-same-candidates, different-rubric-different-gold group.

Family	Type	Rows	In rubric groups
policy_permit	Noul	9,000	5,205
adequacy	Noul	5,000	0
support	Noul	6,000	0
fact	Noul	5,000	0
eligibility*	Noul	1,500	851
routing	Choice	9,003	5,208
action_select	Choice	7,000	4,054
enum_extract	Choice	7,000	3,618
categorical	Choice	7,004	3,882
tool_select*	Choice	1,500	905
quality	Score	6,000	0
severity	Score	6,000	3,497
relevance	Score	3,998	0
completeness	Score	4,001	2,196
urgency*	Score	1,502	900

Table 14: **data_wh** (DecisionMix v2): 60,605 train rows across 12 domains $\times$ 6 boolean conditions, levels 1–6; level 7 (temporal/numeric/probability/trade-off composition) is eval-only.

Level	Train rows	Held-out grammar rows	Eval file	Rows
1	10,601	142	wh_heldout_family	422
2	10,600	141	wh_heldout_style	403
3	10,600	141	wh_heldout_grammar	816
4	10,601	140	wh_level7	880
5	10,602	142	wh_rubric_flip	2,206
6	10,601	141	wh_rubric_shuffled	2,437

A.9 Hard compositional tier and the frozen 14B comparison

Compositions deeper than anything in training are the frontier of single-pass readout, and this subsection collects the measurements that locate it. Rubric-conditioned training moves held-out families, grammars and styles inside the generator’s distribution to .84–.90 (Appendix A.6). Level-7 composition, which requires temporal arithmetic, unit conversion, expected-value reasoning or multi-

Table 15: data_u: 31,029 train rows, 100% soft-target.

Source	Train rows
Synthetic (4 closed-form generators)	28,560
UNLI (validation split only)	2,068
AmbiEnt (ambiguous rows)	401
By decision type: Noul / Score / Choice	23,403 / 7,225 / 401

attribute trade-offs, moves far less. On the 880-item level-7 set the uniform-guess rate is .441 (598 items at $K=2$, 224 at 3, 58 at 4; the majority-class rate is .250), and the checkpoints score .493 (ts1b), .544 (tm1b) and .620 (t11b). The 14B checkpoint is the one clearly above the guess rate, and all three sit 25–40 points below their own in-distribution held-out scores: the curriculum that moves every in-distribution family by more than 30 points moves level 7 by 5. The JevBench hard tier’s temporal, probability and trade-off families show the same pattern.

Table 16: 14B comparisons. **JevBench: public 231-id subset, unranked**; 95% cluster-bootstrap intervals over the paraphrase group field ($n_{\text{eff}}=36$ standard states, $n=111$ hard items). t11b_nokd is identical to t11b in every flag except `-distill_beta 0`. Paired per-item tests are in the text; the marginal intervals overlap and are not the basis for any comparison.

	ladder_14b	t11b	t11b_nokd	Frozen, 3-shot
JevBench std	.875 [.792, .944]	.931 [.861, .986]	.917 [.833, .986]	.819 [.708, .917]
JevBench hard	.468 [.378, .559]	.450 [.360, .541]	.477 [.387, .568]	.559 [.468, .649]
JevBench Brier std (hard)	.173 (.85)	.175 (.66)	.127 (–)	.30 (.60)
long_policy (hard subfamily)	.053	.158	.211	.421
Training val NLL	–	.438	.410	–
Held-out score NLL	2.87	0.95	–	–
Level-7 composition (guess rate .441)	–	.620	–	–

The frozen 14B backbone with three exemplars leads the trained 14B on the hard tier. Table 16 compares the previous-recipe 14B checkpoint (ladder_14b); the bundle that fixed the long-state rendering, dropped over-length rows, and added DecisionMix v2, the typed heads and calibration-shaped checkpoint selection (t11b); the same bundle without frozen-teacher distillation (t11b_nokd); and the frozen backbone with three exemplars. A paired cluster bootstrap on the per-item difference over the same 111 hard items gives frozen-minus-trained $+ .108$ [$+ .027$, $+ .189$], $p=.015$ against t11b, and $+ .081$ [$-.009$, $+ .171$], $p=.082$ against t11b_nokd. On the standard tier the ordering reverses (frozen-minus-trained $-.111$ [$-.250$, $+ .014$], $p=.10$). The frozen control is scored with a 3-shot letter-logit rendering that was not swept; rendering alone moves a frozen 9B checkpoint by 12.5 standard-tier points (Table 7).

The backbone carries hard-tier behavior that the current recipe does not preserve. Per family (Figure 8, Table 18), the frozen 14B leads t11b on tradeoff (.500 vs. .167), long_policy (.421 vs. .158), ambiguous (.714 vs. .429), probability (.500 vs. .300) and temporal_numeric (.333 vs. .200), and trails on multi_hop (.444 vs. .611). Families hold 3–19 items and no single family separates; the direction is the same across every serial family. Our generators compose boolean rules up to six levels deep and never produce the temporal, probability and trade-off operations these families use, which explains why training did not add the behavior. The measurement that separates missing coverage from a limit of fixed-depth computation is the same frozen backbone given a scratchpad on the same 111 items; Section 8 takes this up.

Same-backbone frozen-teacher distillation has no detectable effect. It is the one component of the 14B bundle with its own matched control. t11b_nokd is nominally ahead on hard accuracy (.477 vs. .450), long_policy (.211 vs. .158), training validation NLL (.410 vs. .438) and standard-tier Brier (.127 vs. .175), and t11b is nominally ahead on standard accuracy (.931 vs. .917). The paired bootstraps are $-.027$ [$-.090$, $+ .027$] ($p=.45$) on hard and $+ .014$ [$-.042$, $+ .069$] ($p=.81$) on standard.

The bundle’s clear effect is on probability quality. The ladder_14b $\rightarrow$ t11b standard-tier change is .875 $\rightarrow$.931 (paired $+.056$ [$+.000, +.125$], $p=.13$). Held-out score NLL falls from 2.87 to 0.95 and hard-tier Brier from .85 to .66, both large relative to their noise.

The same question on 19 JevBench items. ladder_14b scores .053 on the JevBench long_policy family ($n=19$) and .919 on the 605 held-out long states in its own training render (Table 6). Before the 2×2 of Section 7.6, the facts-first vs. facts-last question was asked on JevBench (Table 17). The two 1.7B arms are identical in every flag except whether data_wf_long renders its facts at 0% or 97.6% of the state text (same rows, same labels, matched state length, $p50 = 8,702$ characters in both arms), trained at `-max_state 1024` with `-drop_truncated` off so that truncation bites exactly as it did before the fix. On 19 long_policy items in one fixed benchmark render the arms read .316 (6/19) vs. .105 (2/19), Fisher exact $p=.232$, with the hard-tier aggregate moving the other way. On 605 items under the 2×2 , which separates the training effect from render compatibility, the same arms give $+.112$ [$+.077, +.147$], $p = 5 \times 10^{-10}$, and the 14B pair gives $+.078$ [$+.055, +.100$], $p = 7 \times 10^{-12}$ (Table 6).

Table 17: Matched facts-first vs. facts-last training arms, 1.7B, single variable, scored on JevBench: the question of Table 6 asked on 19 long_policy items in a single fixed render. **Public 231-id subset, unranked**, 95% cluster-bootstrap intervals.

	Arm A (facts-first)	Arm B (facts-last)	Test			
long_policy	.316 (6/19)	.105 (2/19)	Fisher	exact	$p=.232$;	bootstrap Δ
JevBench hard	.378 [.288, .468]	.396 [.306, .486]	[$-.053, +.474$]			
JevBench std	.750 [.625, .861]	.736 [.597, .861]	opposite direction indistinguishable			

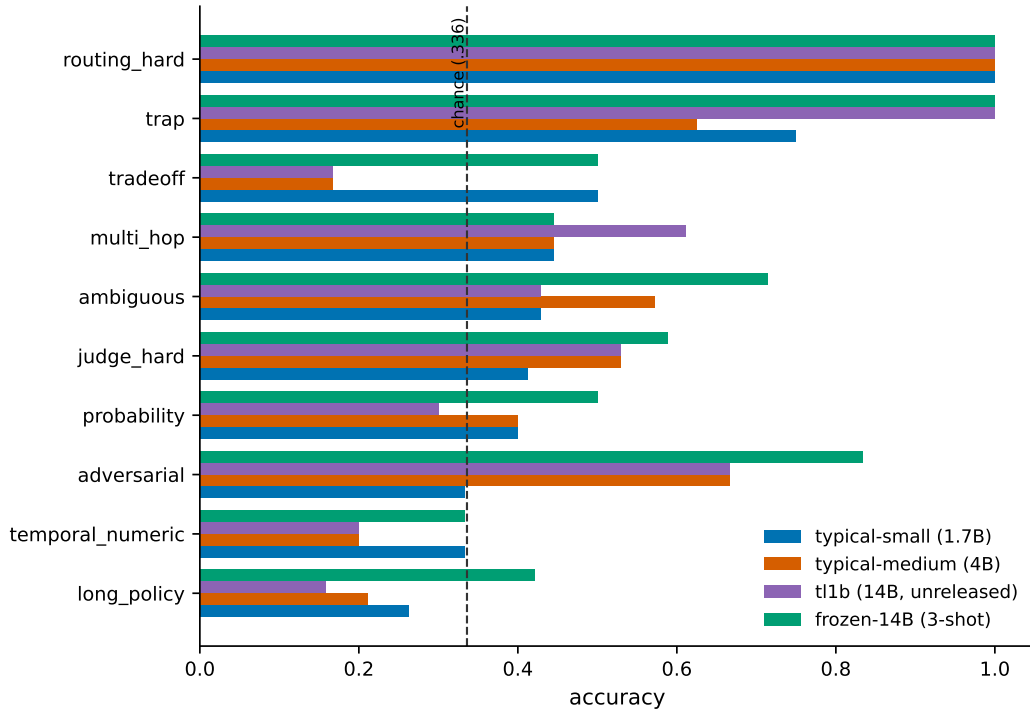


Figure 8: **Public subset, unranked.** Per-family JevBench-hard accuracy for the v1 release checkpoints (ts1b, tm1b), the 14B checkpoint t11b, and the frozen-14B 3-shot control, chance line at .336. Per-family n ranges from 4 to 19 items; no single family separates, and the direction across the serial families is uniform. Values are read directly from each run’s JSON evaluation artifact.

A.9.1 Per-family JevBench-hard table

Table 18 gives the hard-tier sub-families at 14B. The frozen-control column and the ambiguous row are read from each run’s JSON evaluation artifact. For ladder_14b’s temporal and probability cells, the JSON artifact and the project’s internal report disagree (.333/.50 in the JSON vs. .20/.40 in the report); the table gives the report values, marked †.

Table 18: JevBench hard-tier sub-family accuracy, 14B. **Public subset, unranked**; per-family cells are 3–19 items each and carry no useful interval.

Sub-family	ladder_14b	t11b	Frozen 14B, 3-shot
long_policy	.05	.158	.421
multi_hop	.44	.611	.444
trap	1.00	1.00	1.00
adversarial	.67	.667	.833
ambiguous	.429	.429	.714
temporal	.20 [†]	.20	.333
probability	.40 [†]	.30	.50
tradeoff	.67	.17	.50

† The JSON-derived values for these two ladder_14b cells are .333 (temporal) and .50 (probability).

A.10 Closed-loop decisions: a case study

We drove three games with typical-small through the public decision API, one decision per tick, on a laptop (Apple M4 Pro, MPS; the probes ran against the v1 weights, before the v2 upload). Each tick the game engine renders the current situation as the state, offers the legal actions as the candidate set, and executes the argmax. The state holds only the situations that currently apply, each written as a sentence that already contains the comparison a rule needs. Drive and Doom put a precedence-ordered list of one-condition rules in the question; Snake encodes its rules in the state sentences and asks a plain question. Accuracy is agreement with the literal rule list applied over the offered actions (Table 19).

Table 19: Closed-loop decisions with typical-small. “Probe” is agreement with the literal rule list on scripted situations; “chance” is uniform over the offered legal actions; “no rules” keeps the state and replaces the rule list with a goal question.

Game	Candidates (legal ac- tions)	Probe	Chance	No rules	Recorded run
Drive	7 manoeuvres, lane changes only when legal	70/70	.16	.157	231/231 ticks match the rules; 18/18 red lights, 3 lane changes, exit taken
Doom (aiming)	retreat, shoot, turn left, turn right, explore; no shoot at 0 ammo	48/48	.20	.729	arena 240/240, 8 kills; E1M1 200/200 (133 explore, 64 shoot, 3 turn), 4 kills
Snake	safe moves	.984	.48	–	216/217 ticks, 24 food

Drive. The rule list is stop, brake, change lane right, accelerate, change lane left, follow, and a hold speed default, with lane changes removed from the candidate set when the lane is occupied or leads away from the exit. The model matched the rules on 70 scripted situations, including 14/14 exit lane changes and 15/15 overtakes. Removing the guardrail and offering all seven actions drops agreement to .771.

Doom. The model picks one of five actions and the engine presses the key, with a closed-loop turn by the enemy’s bearing. On 48 scripted situations it matched the rules 48/48, including 29/29 turns, and it matched 20/20 sentences in the real engine’s vocabulary. In a recorded run on DOOM’s E1M1, with the engine walking a scripted route when the model chooses explore, all 200 decisions match the rule list: 133 explore, 64 shoot and 3 turns, for 4 kills.

Snake. The state is sentences such as “The snake must move right to close the larger gap to the food.” and “Moving left hits the snake’s body.”, the question is “Which move brings the snake closer to the

food without dying?”, and the candidates are the safe moves. It matched the literal greedy policy on 61 of 62 boards (.984). Moving the same rules into the question as four direction rules without a default scored .82–.95 with $P(\emptyset)$.84–.94.

What the rendering controls. The same weights and the same rules succeed or fail depending on how the state is written, which makes these games a closed-loop instance of Section 7.7. Four regularities held across games.

- *Agent-subject sentences that pre-compute the comparison fire their rule.* “The player must turn left to face the enemy.” fires `turn left` 29/29; the object-location form “The nearest enemy is 5 cells to the left.” fired 0/12 over five phrasings. In Drive, “The car must take the exit on the right in 200 m.” fires 11/11, and “The exit on the right is 200 m ahead.” 0/11. In Snake, the pre-computed sentences score .984 against .774 for the coordinate form “The food is 3 cells to the right and 2 cells up.” on the same 62 boards.
- *An explicit default rule beats “none of the above”.* “explore: applies when the player sees no enemy” fires 6/6; “explore: applies when none of the above rules fire” fires 0/8, with the mass going to shoot and retreat.
- *Raw numbers the model would have to compare leak into the decision.* With “The player has 41 health and 34 ammo.” in the state, an imp in the crosshair reads `retreat .56 / shoot .42`; without the sentence `shoot` is .89. Mid-health crosshair situations score .500 with the numbers sentence and 1.000 without it, and 66 of 200 decisions in an earlier recorded run failed this way. Health and ammo now live on the game display, outside the state.
- *One condition per rule, within the trained list length.* A rule containing “or” fires everywhere (Drive .30 and .61 with a disjunctive rule vs. 1.00 without). A rule at position 6 fired 0/5 under four phrasings and the same rule at position 4 fired 4–5/5, matching the training shape of three rules plus a default.

A.11 Development chronology

Sections 4, 6 and 7 are organized by claim. The experiments ran in this order: (1) a candidate-blind decision-state branch (dual-encoder-style $Z(x, q)$ with an energy scorer over cached candidate embeddings), which produced the ladder of Table 4 and was closed by the shuffled-question control; (2) a candidate-in-suffix decision pass, comparing letter-logit, candidate-blind semantic, and contextual-candidate readouts, which produced Table 3; (3) the last-layer tap and rendered abstain-line diagnoses, which merged a planned two-regime (energy + native) architecture into a single model; (4) the workflow and uncertainty corpora, the mixture sweep, and null augmentation; (5) the Qwen3 size ladder at 1.7B/4B/8B/14B with frozen controls; (6) the 14B bundle (facts-first long-state rendering, `-drop_truncated`, DecisionMix v2, typed heads, calibration-shaped checkpoint selection, frozen-teacher KD) and its matched no-KD control; (7) the Qwen3.5 port; (8) cluster bootstraps, the matched truncation ablation, and the purpose-built held-out long-state set, which produced the paired tests of Appendix A.9 and the 2×2 of Section 7.6; and (9) the v2 releases, `ts1c` (typical-small) and `tm2` (typical-medium), both trained on the facts-first corpus.

The candidate-blind branch was closed on Δ_q . Each student closed 20–55% of the chance-to-teacher accuracy gap, and the shuffled-question control showed that none of the closed gap was question-dependent. Selection on accuracy would have kept that branch.

A.12 JevBench leaderboard context

Table 20 places our numbers among other entries scored on the same 231 public ids. Entries differ in protocol and rendering, several have no method paper, and the 146 private judge items are not included. The public standard tier has $n_{\text{eff}}=36$ independent states, and rendering alone moves a frozen model by 12.5 points (Section 7.7), which exceeds most of the spread in this table (Section 5.5).

A.13 Long-state truncation: distributions and the held-out evaluation set

Figure 5 gives the `data_wf_long` state-length histogram with truncation fractions at 256/1024/2048/3072 tokens, which is the measurement behind the 98.8% figure of Section 4.5, together with `long_policy` accuracy across the bug/fix sequence. The tokenizer’s `truncation_side` is ‘right’, verified empirically on the released tokenizer.

Table 20: Other entries scored on the same public 231 ids, from the harness’s own per-item files. **Public subset, unranked**, for all rows including ours. Majority/chance baselines: .311 standard, .284 easy, .336 hard. 95% cluster-bootstrap intervals for our own rows are in Tables 5 and 16; differences smaller than ± 6 –13 points (standard) or ± 9 points (hard) are not rankings.

Entry (backbone)	standard (72)	easy (48)	hard (111)
open-jev-deberta-v3-large (classifier)	.431	1.00	.378
GLiNER2	.639	1.00	.369
jeff (GLiFormer 400M)	.750	1.00	.387
Laya (ModernBERT)	.694	1.00	.351
open-alternative-jev (Qwen3.5-4B)	.833	1.00	.568
system-one-open (Gemma E2B LoRA)	.931	1.00	.486
SemIf (Qwen3.5-4B)	.986	1.00	.613
OpenJev (26B-A4B)	.972	1.00	.640
Jev 1.13.0 (closed)	.986	1.00	.730
Typical, Qwen3-14B (t11b)	.931	1.00	.450
Typical, Qwen3.5-4B (tm2, typical-medium v2)	.861	1.00	.495

The held-out long-state evaluation set (`scripts/make_long_eval.py`, $n=605$) was built for the 2×2 of Table 6. It has zero exact-state overlap with either training corpus, with Case-stem overlaps dropped, and is scored at a 4,096-token window against states with $p50 = 1,965$ and $\max = 2,843$ tokens, so no item is truncated at evaluation in any of the eight cells. The two renderings differ only in whether the evidence block precedes or follows the request text; the items, labels, and candidate sets are identical. The set shares a generator family with the training data, and it measures the training defect of Section 7.6. Significance for the matched-vs-matched comparison is a two-proportion test over the 605 paired items ($z = 6.2$, $p = 5 \times 10^{-10}$ at 1.7B; $p = 7 \times 10^{-12}$ at 14B).

Released checkpoints. The v1 releases were trained before the facts-first fix, and the v2 releases replace them with post-fix checkpoints (Table 21). On facts-first long states `ts1c` gains 34.9 points over `ts1b` and `tm2` gains 33.8 over `tm1b`, while facts-last accuracy moves by -0.8 and $+1.3$. JevBench does not register the change: `ts1c` reads .708/.432 against `ts1b`’s .694/.432.

Table 21: Release checkpoints on the 605 held-out long states (4,096-token window, no truncation), identical items in both render columns. Majority-class floors: .334 overall, .612 on `policy_permit`. JevBench is the public 231-id subset, unranked.

Release	Checkpoint	facts-first	facts-last	policy_permit (first)	JevBench std / hard
typical-small v1	ts1b	.598	.798	.536	.694 / .432
typical-small v2	ts1c	.947	.790	.948	.708 / .432
typical-medium v1	tm1b	.612	.866	.555	.806 / .423
typical-medium v2	tm2	.950	.879	.967	.861 / .495

A.14 Operations and reproducibility notes

Training and evaluation ran on rented RunPod H100 instances (SXM and NVL variants); the infrastructure issues that affected timing are recorded here for reproducibility. A pod created with an SSH-only startup flag never reached a ready state on the account used and billed regardless (encountered repeatedly before being diagnosed); using a fixed exposed port at boot avoided it. Stopping a pod without an attached persistent volume wipes it, so checkpoints and manifests were always written to a network volume or pushed to the Hugging Face Hub before a pod was stopped. `tmux kill-session` does not terminate a `timeout`-wrapped child process, which once produced a stray duplicate training run sharing a GPU with the intended one; process matching over SSH must target the exact command, since a loose `pkill` pattern can match and kill the SSH shell itself before the target process. Two pods sharing one Python environment on the same network volume corrupted it under concurrent dependency resolution; per-pod environments with dependency syncing disabled avoided this. Every run’s manifest records the harness commit, checkpoint hash, and dependency

lock file used, and the JevBench per-item comparisons in this paper are all computed against the same fixed set of 231 public ids from one harness commit.

A.15 Full result tables

Tables 22 to 26 give the complete cross-checkpoint result tables for every checkpoint with a completed evaluation. `ts1b` and `tm1b` are the v1 releases; `tm2` is typical-medium v2, and `ts1c` (typical-small v2) is in Table 21.

Table 22: Evidence and knowledge-classification accuracy by checkpoint.

Model	Checkpoint	CLINC-150	TREC-fine	HWU64	20NG	SNLI	MNLI	BoolQ	ANLI
Qwen3-1.7B	<code>ts1b</code>	.801	.508	.761	.540	.894	.859	.824	.497
Qwen3-4B	<code>tm1b</code>	.847	.414	.769	.588	.909	.861	.843	.544
Qwen3-8B	<code>ladder_8b</code>	.769	.486	.740	.452	.853	.831	.822	.468
Qwen3-14B	<code>t11b</code>	.827	.508	.760	.690	.911	.868	.882	.583
Qwen3.5-4B	<code>tm2</code>	.795	.480	.717	.579	.900	.855	.840	.554

Table 23: Knowledge probe (MMLU-Pro among- K , Δ_q^{sh}) and TruthfulQA MC1.

Model	Checkpoint	MMLU-Pro among- K	Δ_q^{sh}	TruthfulQA MC1
Qwen3-1.7B	<code>ts1b</code>	.343	.127	.257
Qwen3-4B	<code>tm1b</code>	.458	.193	.408
Qwen3-8B	<code>ladder_8b</code>	.302	.093	.277
Qwen3-14B	<code>t11b</code>	.498	.235	.481
Qwen3.5-4B	<code>tm2</code>	.429	.194	.414

Table 24: Held-out workflow and external floors. Rows marked *full* are the post-hoc `eval_wf` pass over every row at full state length (4,096-token window, `-limit 0`); rows marked *train-time* are the 1,500-row, 256-token pass emitted during training, which truncates the long external sets and is not comparable to the full pass. The distinction is load-bearing on the external sets: on `ts1b`, PagerDuty reads .560 at train time and .817 at full length, because the 256-token window removes the incident text. Numbers in parentheses after held-out score are NLL.

Model	Pass	Held-out noul	Held-out score (NLL)	Held-out style	PagerDuty (.792)	jevlogs (.697)	Mind2Web (.427)
<code>ts1b</code>	<i>full</i>	.715	.520 (1.01)	.859	.817	.710	.357
<code>tm1b</code>	<i>full</i>	.811	.528 (1.03)	.859	.838	.673	.544
<code>t11b</code>	<i>full</i>	.831	.623 (0.95)	.880	.878	.546	.563
<code>ladder_8b</code>	<i>train-time</i>	.681	.543 (1.72)	.882	.321	.501	.404
<code>tm2</code>	<i>train-time</i>	.858	.586 (0.97)	.842	.757	.518	.497

Table 25: JevBench accuracy (Brier). **Public subset, unranked** (72 standard over 36 paraphrase-clustered states / 48 easy / 111 hard); see Tables 5 and 16 for intervals.

Model (checkpoint)	Standard (Brier)	Easy (Brier)	Hard (Brier)
Qwen3-1.7B (ts1b)	.694 (.40)	1.00 (.005)	.432 (.79)
Qwen3-4B (tm1b)	.806 (.30)	1.00 (.012)	.423 (.77)
Qwen3-8B (ladder_8b)	.667 (.55)	1.00 (.015)	.396 (.83)
Qwen3-14B (t11b)	.931 (.18)	1.00 (.005)	.450 (.66)
Qwen3.5-4B (tm2)	.861 (.22)	1.00	.495 (.72)
Qwen3.5-4B, frozen 3-shot	.764 (.36)	1.00	.495 (.60)
Qwen3.5-9B, frozen semif-render	.931 (.11)	1.00	.595 (.51)
Qwen3-14B, frozen 3-shot	.819 (.28)	1.00	.559 (.56)

Table 26: Single-decision latency (same-pod, same-torch-build; ms), $L_s=256$.

Model (checkpoint)	$K=2$	$K=32$	$K=256$	Peak memory $K=2 \rightarrow 256$ (GB)
Qwen3-1.7B (ts1b-shape)	45	46	106	6.4 $\rightarrow$ 9.3
Qwen3-4B (tm1b)	57	58	96	14.6 $\rightarrow$ 18.6
Qwen3-8B (ladder_8b)	70	71	126	29.3 $\rightarrow$ 34.1
Qwen3-14B (t11b)	59	62	156	51.6 $\rightarrow$ 57.5